```
library(margins)
library(modelr)
library(patchwork)
library(pbkrtest)
library(psych)
library(purrr)
library(pwr)
library(readr)
library(readxl)
library(rstanarm)
library(scales)
library(semPlot)
library(stringr)
library(tidyLPA)
library(tidyr)
library(tidyverse)
```

Data

```
# adding data set
data <- read.csv("data.csv")

data$sex <- as.factor(data$sex)
data$ethnicity <- as.factor(data$ethnicity)
data$year <- as.factor(data$year)

str(data)
```

```
'data.frame':   390 obs. of  376 variables:
 $ N                     : int  1 2 3 4 5 8 9 10 12 13 ...
 $ Progress              : int  100 100 100 100 100 100 100 100 100 100 ...
 $ Duration..in.seconds.: int  1415 3058 1550 1895 1068 1457 1001 43652 1350 11984 ...
 $ LS1                   : chr  "QPPQ" "DDDE" "QPPQ" "QPPQ" ...
 $ LS2                   : chr  "PRST" "BCDE" "PRST" "PRST" ...
 $ LS3                   : chr  "PWXQ" "JVZK" "PWXQ" "PWXQ" ...
 $ LS4                   : chr  "BCEF" "CDFG" "STWX" "STWX" ...
 $ LS5                   : chr  "WXXW" "LMML" "GFFG" "GFFG" ...
 $ LS6                   : chr  "CCRR" "AAPP" "QQBB" "QQBB" ...
 $ LS7                   : chr  "EGFH" "ABDC" "IJLK" "EGFH" ...
```

```
$ LS8                : chr  "BODQ" "CERT" "FHXZ" "BODQ" ...
$ LS9                : chr  "SEFT" "SEFT" "COPD" "FUZG" ...
$ LS10               : chr  "JCVC" "CGCS" "CGCS" "CLXC" ...
$ LS11               : chr  "VEGV" "VEGV" "XDBK" "XDBK" ...
$ LS12               : chr  "CEIZ" "CEIZ" "CGVZ" "CGVZ" ...
$ LS13               : chr  "VEBT" "VEBT" "VEBT" "VEBT" ...
$ LS14               : chr  "EJFK" "PSQT" "EJFK" "GKHM" ...
$ LS15               : chr  "LHEC" "HJMQ" "HJMQ" "HJMQ" ...
$ FC1_1              : chr  "5,3,2" "8,7,6,4,1" "3,5,7" "3,5,2" ...
$ FC1_2              : chr  "8,7,6,4,1" "5,3,2" "6,2,1" "8,7,6,4,1" ...
$ FC2_1              : chr  "6,4,2" "7,4,2,3,6" "2,3,1" "6,4,2" ...
$ FC2_2              : chr  "8,7,5,3,1" "8,5,1" "6,5,4" "8,7,5,3,1" ...
$ FC3_1              : chr  "7,4,1" "6,4,3,2" "3,2,1" "7,4,1" ...
$ FC3_2              : chr  "8,6,5,3,2" "7,8,5,1" "5,7,4" "8,6,5,3,2" ...
$ FC4_1              : chr  "8,6,3,2" "7,5,4,1" "4,2,3" "8,6,3,2" ...
$ FC4_2              : chr  "7,5,4,1" "6,3,8,2" "8,7,1" "7,5,4,1" ...
$ FC5_1              : chr  "6,8,3,1" "8,7,4,5,3,2" "2,6,4" "8,5,4,2" ...
$ FC5_2              : chr  "5,7,4,2" "6,1" "7,5,1" "7,6,3,1" ...
$ FC6_1              : chr  "7,6,4,1" "2,5,4,1" "3,1,4" "6,4,8,5,3,2,1" ...
$ FC6_2              : chr  "8,5,3,2" "8,7,6,3" "7,6,2" "7" ...
$ FC7_1              : chr  "8,6,3,1" "8,6,5,3,1" "2,8,1" "7,5,3,8,2" ...
$ FC7_2              : chr  "7,5,4,2" "7,4,2" "4,6,5" "6,4,1" ...
$ FC8_1              : chr  "6,5,3,1" "8,5,2,4" "5,3,1" "6,5,4,1" ...
$ FC8_2              : chr  "8,7,4,2" "7,6,3,1" "8,6,7" "8,7,3,2" ...
$ FC9_1              : chr  "2,1" "6,4,1" "5,1,2" "6,2" ...
$ FC9_2              : chr  "4,3,8" "7,3,2" "7,6,4" "5,4,3" ...
$ FC9_3              : chr  "5,7,6" "8,5" "3,8" "8,7,1" ...
$ FC10_1             : chr  "8,3,1" "7,8,3,1" "3,5,1" "3,1" ...
$ FC10_2             : chr  "5,6,2" "5,4" "7,6,4" "7,4" ...
$ FC10_3             : chr  "7,4" "2,6" "8,2" "8,5,6,2" ...
$ FC11_1             : chr  "8,1,3" "6,4" NA "4,6" ...
$ FC11_2             : chr  "6,4,2" "3,1,5" NA "5,1,3" ...
$ FC11_3             : chr  "7,5" "2,8,7" NA "2,7,8" ...
$ FC12_1             : chr  "1,5,6" "4,1,2" NA "2,5,7" ...
$ FC12_2             : chr  "2,8,4" "3,6,8" NA "1,4,3" ...
$ FC12_3             : chr  "3,7" "5,7" NA "8,6" ...
$ FC13_1             : chr  "2,5,1" "1,4,6" NA "1,8,7" ...
$ FC13_2             : chr  "3,6,4" "2,8" NA "2,6" ...
$ FC13_3             : chr  "7,8" "3,7,5" NA "3,4,5" ...
$ FC14_1             : chr  "1,3,6,8" "5,2,3" NA "2,5" ...
$ FC14_2             : chr  "2,5,4" "7,8" NA "3,7,8" ...
$ FC14_3             : chr  "7" "1,4,6" NA "1,6,4" ...
$ CN_1_1             : int  3 5 4 5 6 6 6 6 6 5 ...
```

```
$ CN_1_2             : int  5 5 5 4 6 5 6 6 7 5 ...
$ CN_1_3             : int  5 5 5 5 6 6 5 5 7 5 ...
$ CN_1_4             : int  4 5 5 3 6 6 1 6 7 5 ...
$ CN_1_5             : int  4 6 5 6 6 6 5 6 7 5 ...
$ CN_1_6             : int  3 5 5 6 6 6 5 5 6 5 ...
$ CN_1_7             : int  5 5 5 5 6 4 4 7 6 5 ...
$ CN_1_8             : int  3 5 5 5 6 6 5 5 6 5 ...
$ CN_2_1             : int  5 6 4 4 6 5 6 5 4 6 ...
$ CN_2_2             : int  5 5 5 5 6 5 5 6 6 6 ...
$ CN_2_3             : int  5 5 5 4 6 4 6 7 6 5 ...
$ CN_2_4             : int  4 5 5 4 6 6 5 7 5 5 ...
$ CN_2_5             : int  5 5 5 4 6 6 5 7 6 6 ...
$ CN_2_6             : int  3 5 5 5 6 6 6 7 6 6 ...
$ CN_2_7             : int  3 5 5 5 5 5 5 7 6 5 ...
$ CN_2_8             : int  5 5 5 5 6 4 5 7 6 5 ...
$ CN_3_1             : int  4 5 5 5 6 5 6 3 5 6 ...
$ CN_3_2             : int  4 5 5 6 6 6 6 4 6 4 ...
$ CN_3_3             : int  4 5 5 6 6 5 5 5 5 5 ...
$ CN_3_4             : int  5 4 5 5 6 4 5 4 6 4 ...
$ CN_3_5             : int  6 7 5 7 7 4 5 7 5 4 ...
$ CN_3_6             : int  5 6 5 7 7 5 6 7 5 4 ...
$ CN_3_7             : int  3 6 5 5 7 4 7 4 5 6 ...
$ CN_3_8             : int  3 6 6 7 7 5 6 6 5 5 ...
$ CN_4_1             : int  7 7 5 6 6 5 5 6 5 5 ...
$ CN_4_2             : int  7 7 4 6 7 5 6 5 5 4 ...
$ CN_4_3             : int  6 7 4 7 6 5 5 5 5 4 ...
$ CN_4_4             : int  7 7 5 7 7 5 5 6 6 5 ...
$ XXX1               : int  3 3 3 3 3 3 3 3 3 3 ...
$ CN_4_5             : int  7 7 5 6 7 5 5 6 6 4 ...
$ CN_4_6             : int  6 7 4 6 7 5 5 6 5 5 ...
$ CN_4_7             : int  7 5 5 6 6 5 5 7 6 5 ...
$ CN_4_8             : int  7 7 5 6 7 5 5 7 5 6 ...
$ CN_5_1             : int  4 7 4 7 7 4 6 6 7 4 ...
$ CN_5_2             : int  3 6 4 5 7 4 6 7 5 4 ...
$ CN_5_3             : int  3 7 5 7 7 5 6 5 5 5 ...
$ CN_5_4             : int  3 7 4 6 6 4 7 5 6 4 ...
$ CN_5_5             : int  4 7 4 7 7 4 5 7 5 5 ...
$ CN_5_6             : int  3 7 5 6 7 5 5 5 5 3 ...
$ CN_5_7             : int  5 7 5 7 7 4 7 6 5 5 ...
$ CN_5_8             : int  4 7 5 5 7 5 6 7 5 4 ...
$ MA_1_1             : int  7 7 6 6 7 6 5 4 7 5 ...
$ MA_1_2             : int  6 6 6 7 7 6 5 5 6 5 ...
$ MA_1_3             : int  5 7 6 4 7 6 4 5 6 6 ...
```

```
 $ MA_1_4               : int  4 7 7 7 7 6 5 7 6 6 ...
 $ MA_1_5               : int  4 4 4 3 7 5 3 3 6 3 ...
 $ MA_1_6               : int  7 7 6 6 7 7 5 6 7 6 ...
  [list output truncated]
```

Descrip

```
age_summary<-data|>
  summarize(Variable= "Age", Mean=mean(age), Median=median(age),sd= sd(age), n= n())
```

```
descriptives <- describe(data[, c(
  "age", "gpa", "act", "sat_reading", "sat_math",
  "LS", "FC", "EQ",
  "CN_S", "CN_A", "CN_I", "CN_C", "CN_M",
  "M_S", "M_B", "M_Se", "M_Sa",
  "EI_Ch", "EI_Cu", "EI_I", "EI_E", "EI_P", "EI_D",
  "MC_S", "MC_P", "MC_A",
  "NE_N", "NE_O", "NE_C",
  "AR_1_1", "AR_1_2", "AR_1_3", "AR_1_4", "AR_1_5",
  "AR_2", "AR_3",
  "AI_1", "AI_2", "AI_3", "AI_4", "AI_5"
)])

descriptives <- descriptives[, c("n", "mean", "median", "sd", "min", "max")]

descriptives |>  kable (caption= "Summary Statistics",digits=2)
```

Table 1: Summary Statistics

	n	mean	median	sd	min	max
age	390	19.93	20.00	1.44	18	30.00
gpa	354	3.65	3.73	0.37	2	4.25
act	125	32.88	34.00	3.39	13	36.00
sat_reading	230	727.08	740.00	57.03	370	800.00
sat_math	230	750.41	770.00	60.47	420	800.00
LS	386	9.93	11.00	3.47	1	15.00
FC	389	6.16	6.00	2.93	0	14.00
EQ	390	169.27	170.00	20.89	112	228.00
CN_S	390	44.47	44.00	6.56	22	56.00

	n	mean	median	sd	min	max
CN_A	390	41.57	42.00	7.88	19	56.00
CN_I	390	41.60	41.00	6.99	18	56.00
CN_C	390	45.67	46.00	6.39	30	56.00
CN_M	390	41.38	42.00	9.46	11	56.00
M_S	390	51.68	52.00	8.88	23	70.00
M_B	390	59.54	61.00	8.57	29	70.00
M_Se	390	57.71	58.00	8.46	32	70.00
M_Sa	390	52.28	52.00	8.06	26	70.00
EI_Ch	390	28.88	29.00	6.74	6	42.00
EI_Cu	390	28.79	29.00	6.43	10	42.00
EI_I	390	24.95	25.00	5.39	6	35.00
EI_E	390	20.18	21.00	5.82	5	35.00
EI_P	390	13.85	14.50	3.79	3	21.00
EI_D	390	20.46	21.00	6.11	5	35.00
MC_S	390	22.81	23.00	3.21	10	28.00
MC_P	390	13.02	13.00	3.44	3	21.00
MC_A	390	15.67	16.00	2.96	5	21.00
NE_N	390	47.15	47.00	11.04	16	77.00
NE_O	390	58.88	59.00	9.24	28	84.00
NE_C	390	60.38	61.00	8.97	29	82.00
AR_1_1	390	5.45	6.00	1.31	1	7.00
AR_1_2	390	4.48	5.00	1.54	1	7.00
AR_1_3	390	4.87	5.00	1.44	1	7.00
AR_1_4	390	5.41	6.00	1.20	1	7.00
AR_1_5	390	5.06	5.00	1.42	1	7.00
AR_2	390	5.26	5.00	1.25	1	7.00
AR_3	364	5.00	5.00	1.30	1	7.00
AI_1	390	5.62	6.00	1.27	1	7.00
AI_2	389	5.14	5.00	1.49	1	7.00
AI_3	390	5.82	6.00	1.19	1	7.00
AI_4	390	5.99	6.00	1.02	1	7.00
AI_5	390	5.93	6.00	1.30	1	7.00

```
categorical_summary <- data |>
  select(sex, ethnicity, year) |>
  pivot_longer(everything(),
               names_to = "Variable",
               values_to = "Category") |>
  filter(!is.na(Category)) |>
  count(Variable, Category) |>
```

```
  group_by(Variable) |>
  mutate(Percentage = round(n / sum(n) * 100, 1))

categorical_summary |>  kable (caption= "Summary Statistics",digits=2)
```

Table 2: Summary Statistics

Variable	Category	n	Percentage
ethnicity	Asian	108	27.8
ethnicity	Asian American	31	8.0
ethnicity	Black or African American	26	6.7
ethnicity	Hispanic or Latino	29	7.5
ethnicity	prefer not to answer	11	2.8
ethnicity	two or more races	45	11.6
ethnicity	White or Caucasian	139	35.7
sex	Female	273	70.0
sex	Male	113	29.0
sex	other	2	0.5
sex	Prefer not to say	2	0.5
year	Fifth year or beyond	4	1.0
year	First year (Freshman)	57	14.7
year	Fourth year (Senior)	99	25.4
year	Second year (Sophomore)	134	34.4
year	Third year (Junior)	95	24.4

Reliability

```
#psych::alpha(data[, paste0("EQ", 1:33)]) |> print ()

# Emotional Intelligence
psych::alpha(data[, paste0("EQ", 1:33)])$total$raw_alpha
```

```
[1] 0.8986268
```

```
# Core Needs
psych::alpha(data[, paste0("CN_1_", 1:8)])$total$raw_alpha   # Security
```

```
[1] 0.8199992
```

```r
psych::alpha(data[, paste0("CN_2_", 1:8)])$total$raw_alpha   # Acceptance
```

```
[1] 0.8866242
```

```r
psych::alpha(data[, paste0("CN_3_", 1:8)])$total$raw_alpha   # Initiative
```

```
[1] 0.8422906
```

```r
psych::alpha(data[, paste0("CN_4_", 1:8)])$total$raw_alpha   # Competence
```

```
[1] 0.8940293
```

```r
psych::alpha(data[, paste0("CN_5_", 1:8)])$total$raw_alpha   # Meaning
```

```
[1] 0.9216608
```

```r
# Maslow
psych::alpha(data[, paste0("MA_1_", 1:10)])$total$raw_alpha  # Safety
```

```
[1] 0.8634134
```

```r
psych::alpha(data[, paste0("MA_2_", 1:10)])$total$raw_alpha  # Belonging
```

```
[1] 0.8818991
```

```r
psych::alpha(data[, paste0("MA_3_", 1:10)])$total$raw_alpha  # Self-esteem
```

```
[1] 0.9214845
```

```r
psych::alpha(data[, paste0("MA_4_", 1:10)])$total$raw_alpha  # Self-actualization
```

```
[1] 0.8233781
```

```r
# Extrinsic / Intrinsic Motives
psych::alpha(data[, paste0("EI_1_", 1:6)])$total$raw_alpha   # Challenge
```

```
[1] 0.8892558
```

```r
psych::alpha(data[, paste0("EI_2_", 1:6)])$total$raw_alpha   # Curiosity
```

```
[1] 0.8173063
```

```r
psych::alpha(data[, paste0("EI_3_", 1:5)])$total$raw_alpha   # Independence Mastery
```

```
[1] 0.868526
```

```r
psych::alpha(data[, paste0("EI_4_", 1:5)])$total$raw_alpha   # Easy Work
```

```
[1] 0.8323405
```

```r
psych::alpha(data[, paste0("EI_5_", 1:3)])$total$raw_alpha   # Pleasing Teacher
```

```
[1] 0.7585386
```

```r
psych::alpha(data[, paste0("EI_6_", 1:5)])$total$raw_alpha   # Dependence on Teacher
```

```
[1] 0.8274458
```

```r
# McClelland
psych::alpha(data[, paste0("MC_1_", 1:4)])$total$raw_alpha   # Self-achievement
```

```
[1] 0.6759667
```

```r
psych::alpha(data[, paste0("MC_2_", 1:3)])$total$raw_alpha   # Power
```

```
[1] 0.6974886
```

```r
psych::alpha(data[, paste0("MC_3_", 1:3)])$total$raw_alpha   # Affiliation
```

```
[1] 0.4760393
```

```r
# NEO
psych::alpha(data[, paste0("NE_1_", 1:12)])$total$raw_alpha  # Neuroticism
```

```
[1] 0.830528
```

```r
psych::alpha(data[, paste0("NE_2_", 1:12)])$total$raw_alpha  # Openness
```

```
[1] 0.7051049
```

```r
psych::alpha(data[, paste0("NE_3_", 1:12)])$total$raw_alpha  # Conscientiousness
```

```
[1] 0.7995361
```

```r
# LS and FC with KR-20

# Letter Sets
psych::alpha(data[, paste0("LS", 1:15, "_score")])$total$raw_alpha
```

```
Warning in psych::alpha(data[, paste0("LS", 1:15, "_score")]): Some items were negatively co
should be reversed.
To do this, run the function again with the 'check.keys=TRUE' option

Some items ( LS15_score ) were negatively correlated with the first principal component and
probably should be reversed.
To do this, run the function again with the 'check.keys=TRUE' option

[1] 0.8183421
```

```r
psych::alpha(data[, paste0("LS", 1:15, "_score")])$item.stats
```

```
Warning in psych::alpha(data[, paste0("LS", 1:15, "_score")]): Some items were negatively co
should be reversed.
To do this, run the function again with the 'check.keys=TRUE' option
```

```
Some items ( LS15_score ) were negatively correlated with the first principal component and
probably should be reversed.
To do this, run the function again with the 'check.keys=TRUE' option
```

	n	raw.r	std.r	r.cor	r.drop	mean	sd
LS1_score	386	0.55031749	0.58441579	0.5537765	0.4792727	0.8782383	0.3274347
LS2_score	386	0.57195716	0.59597121	0.5610893	0.4928572	0.8341969	0.3723865
LS3_score	386	0.58508257	0.62157699	0.5926120	0.5218880	0.8937824	0.3085158
LS4_score	386	0.56022839	0.56499444	0.5210536	0.4646261	0.7461140	0.4357980
LS5_score	386	0.63227502	0.65175819	0.6321893	0.5594917	0.8264249	0.3792353
LS6_score	386	0.58672472	0.59745573	0.5566885	0.5030987	0.8005181	0.4001295
LS7_score	386	0.54222530	0.54286213	0.4889477	0.4424842	0.7331606	0.4428818
LS8_score	386	0.57550849	0.55945269	0.5132882	0.4683309	0.5880829	0.4928190
LS9_score	386	0.61944579	0.60003653	0.5664357	0.5178794	0.4818653	0.5003195
LS10_score	386	0.67295925	0.66627051	0.6436852	0.5883559	0.6787565	0.4675601
LS11_score	386	0.63298029	0.61270548	0.5829123	0.5350700	0.5751295	0.4949649
LS12_score	386	0.59001145	0.58392197	0.5455269	0.4936973	0.7046632	0.4567861
LS13_score	386	0.43771979	0.41453026	0.3430749	0.3104761	0.5284974	0.4998351
LS14_score	386	0.48329082	0.45992451	0.3896164	0.3611877	0.4818653	0.5003195
LS15_score	386	-0.03741213	-0.03141938	-0.1595485	-0.1464866	0.1787565	0.3836455

```r
# Figure Classification
psych::alpha(data[, paste0("FC", 1:14)])$total$raw_alpha
```

```
[1] 0.7955718
```

Correlation

```r
# Load required package
# Select variables for correlations
cor_data <- data[, c(
  "CN_S", "CN_A", "CN_I", "CN_C", "CN_M",
  "M_S", "M_B", "M_Se", "M_Sa",
  "EI_Ch", "EI_Cu", "EI_I", "EI_E", "EI_P", "EI_D",
  "MC_S", "MC_P",
  "NE_N", "NE_O", "NE_C", "EQ",
  "LS", "FC", "gpa")]


variable_labels <- c(
  CN_S  = "CN_Security",
  CN_A  = "CN_Acceptance",
  CN_I  = "CN_Initiative",
  CN_C  = "CN_Competence",
  CN_M  = "CN_Meaning",
  M_S   = "M_Safety",
  M_B   = "M_Belonging",
  M_Se  = "M_Self-esteem",
  M_Sa  = "M_Self-actualization",
  EI_Ch = "EI_Challenge",
  EI_Cu = "EI_Curiosity",
  EI_I  = "EI_Independence Mastery",
  EI_E  = "EI_Easy work",
  EI_P  = "EI_Pleasing",
  EI_D  = "EI_Dependence on teacher",
  MC_S  = "MC_Self-achievement",
  MC_P  = "MC_Power",

  EQ    = "Emotional Intelligence",
  NE_N  = "NE_Neuroticism",
  NE_O  = "NE_Openness",
  NE_C  = "NE_Conscientiousness",
  LS    = "Letter Sets",
  FC    = "Figure Classification",
  NS_S  = "Security satisfaction",
  NS_A  = "Acceptance satisfaction",
  NS_I  = "Initiative satisfaction",
```

```
  NS_C  = "Self-efficacy satisfaction",
  NS_M  = "Meaning satisfaction",
  gpa   = "GPA"
)

names(cor_data) <- unname(variable_labels[names(cor_data)])


intercor_matrix <- cor(cor_data, use = "pairwise.complete.obs", method = "pearson")

print(intercor_matrix)
```

```
                          CN_Security CN_Acceptance CN_Initiative
CN_Security                1.0000000000   0.663484032    0.53300069
CN_Acceptance              0.6634840315   1.000000000    0.55038056
CN_Initiative              0.5330006854   0.550380559    1.00000000
CN_Competence              0.5768268480   0.488480266    0.69712508
CN_Meaning                 0.4821872957   0.467468495    0.61381630
M_Safety                   0.5090872217   0.321604258    0.35718742
M_Belonging                0.5656698827   0.448376915    0.36027607
M_Self-esteem              0.5932860068   0.461417046    0.58801900
M_Self-actualization       0.3792224152   0.425917831    0.58816997
EI_Challenge               0.0801164315   0.151013552    0.35226347
EI_Curiosity               0.2206242515   0.262404321    0.40317234
EI_Independence Mastery    0.1141094091   0.098354285    0.26279391
EI_Easy work               0.0996994300   0.001456188   -0.10349605
EI_Pleasing                0.0717706432   0.013437565   -0.11384500
EI_Dependence on teacher   0.1976054689   0.183910357    0.08053243
MC_Self-achievement        0.2651129756   0.233037992    0.41872634
MC_Power                   0.1086456450   0.076214871    0.18894545
NE_Neuroticism             0.1109826662   0.099468536   -0.03793065
NE_Openness                0.1102219831   0.175860934    0.10573859
NE_Conscientiousness       0.2305387208   0.203111511    0.24726585
Emotional Intelligence     0.2877458214   0.281607300    0.43373623
Letter Sets               -0.0603682099  -0.083451613   -0.12436229
Figure Classification     -0.0547436816  -0.051177969   -0.21344052
GPA                        0.0004787008   0.057566989   -0.09246857
                          CN_Competence   CN_Meaning    M_Safety M_Belonging
CN_Security               0.5768268480  0.482187296  0.50908722  0.56566988
CN_Acceptance             0.4884802657  0.467468495  0.32160426  0.44837691
CN_Initiative             0.6971250820  0.613816296  0.35718742  0.36027607
CN_Competence             1.0000000000  0.597788058  0.44295511  0.41611965
```

CN_Meaning	0.5977880581	1.000000000	0.41816707	0.41255781
M_Safety	0.4429551135	0.418167073	1.00000000	0.39903988
M_Belonging	0.4161196545	0.412557807	0.39903988	1.00000000
M_Self-esteem	0.7198862878	0.581344865	0.52708542	0.61405307
M_Self-actualization	0.4855572253	0.482808618	0.30641109	0.42277839
EI_Challenge	0.2692522559	0.322613268	0.03345958	0.14546967
EI_Curiosity	0.3209193826	0.416982531	0.15493423	0.25179307
EI_Independence Mastery	0.2414139815	0.231000417	0.15297408	0.07820184
EI_Easy work	-0.0421848180	-0.029239573	0.21671306	0.04683165
EI_Pleasing	0.0007547254	-0.042975025	0.15315228	0.05250666
EI_Dependence on teacher	0.1080213021	0.165443073	0.18508170	0.21699956
MC_Self-achievement	0.4529304801	0.375517618	0.19285961	0.38427611
MC_Power	0.2425648281	0.163451827	0.26062418	0.18664104
NE_Neuroticism	0.0055731489	0.003100202	0.18214791	0.01737406
NE_Openness	0.0495968040	0.078001782	-0.12262052	0.11087079
NE_Conscientiousness	0.2836082121	0.296113136	0.24447156	0.30323472
Emotional Intelligence	0.3793600460	0.389222624	0.14966472	0.41307600
Letter Sets	-0.0615562359	-0.173030856	-0.12267344	0.03065208
Figure Classification	-0.1137167377	-0.162965133	-0.15193046	-0.02276039
GPA	-0.0263324090	-0.072777284	-0.08130712	0.10233505

	M_Self-esteem	M_Self-actualization	EI_Challenge
CN_Security	0.593286007	0.37922242	0.08011643
CN_Acceptance	0.461417046	0.42591783	0.15101355
CN_Initiative	0.588018996	0.58816997	0.35226347
CN_Competence	0.719886288	0.48555723	0.26925226
CN_Meaning	0.581344865	0.48280862	0.32261327
M_Safety	0.527085423	0.30641109	0.03345958
M_Belonging	0.614053070	0.42277839	0.14546967
M_Self-esteem	1.000000000	0.61817064	0.23024485
M_Self-actualization	0.618170642	1.00000000	0.27352684
EI_Challenge	0.230244852	0.27352684	1.00000000
EI_Curiosity	0.306148766	0.36352734	0.75882685
EI_Independence Mastery	0.167364821	0.13329756	0.59276670
EI_Easy work	0.033073260	-0.05740910	-0.69140791
EI_Pleasing	0.044558892	-0.03398211	-0.45313859
EI_Dependence on teacher	0.162796918	0.08897508	-0.14490814
MC_Self-achievement	0.479859003	0.44546411	0.58659615
MC_Power	0.303227304	0.18384150	0.06751288
NE_Neuroticism	0.002586548	-0.12015388	-0.25419728
NE_Openness	0.103350041	0.26101722	0.21760015
NE_Conscientiousness	0.280445613	0.18682981	0.32539704
Emotional Intelligence	0.456026741	0.49998446	0.44228244
Letter Sets	-0.030881628	-0.11800527	0.01930366

Figure Classification	-0.091865379	-0.13448426	0.03202353
GPA	-0.029200203	-0.17892702	0.11079319

	EI_Curiosity	EI_Independence Mastery	EI_Easy work
CN_Security	0.22062425	0.11410941	0.099699430
CN_Acceptance	0.26240432	0.09835429	0.001456188
CN_Initiative	0.40317234	0.26279391	-0.103496054
CN_Competence	0.32091938	0.24141398	-0.042184818
CN_Meaning	0.41698253	0.23100042	-0.029239573
M_Safety	0.15493423	0.15297408	0.216713057
M_Belonging	0.25179307	0.07820184	0.046831652
M_Self-esteem	0.30614877	0.16736482	0.033073260
M_Self-actualization	0.36352734	0.13329756	-0.057409099
EI_Challenge	0.75882685	0.59276670	-0.691407911
EI_Curiosity	1.00000000	0.47309479	-0.507426395
EI_Independence Mastery	0.47309479	1.00000000	-0.405823855
EI_Easy work	-0.50742640	-0.40582386	1.000000000
EI_Pleasing	-0.43054442	-0.26478368	0.636395108
EI_Dependence on teacher	0.05674784	-0.30640263	0.290895693
MC_Self-achievement	0.49848352	0.37959811	-0.424420189
MC_Power	0.13497054	0.02914944	0.059892169
NE_Neuroticism	-0.21819675	-0.14142621	0.310802579
NE_Openness	0.23630225	0.05918322	-0.275266105
NE_Conscientiousness	0.30236775	0.26152715	-0.240660464
Emotional Intelligence	0.51134336	0.25570133	-0.263952479
Letter Sets	-0.11490995	0.06468436	-0.073125909
Figure Classification	-0.04457247	0.04283845	-0.146022765
GPA	0.07250894	0.09945813	-0.159062021

	EI_Pleasing	EI_Dependence on teacher
CN_Security	0.0717706432	0.19760547
CN_Acceptance	0.0134375652	0.18391036
CN_Initiative	-0.1138449977	0.08053243
CN_Competence	0.0007547254	0.10802130
CN_Meaning	-0.0429750246	0.16544307
M_Safety	0.1531522816	0.18508170
M_Belonging	0.0525066597	0.21699956
M_Self-esteem	0.0445588919	0.16279692
M_Self-actualization	-0.0339821080	0.08897508
EI_Challenge	-0.4531385858	-0.14490814
EI_Curiosity	-0.4305444184	0.05674784
EI_Independence Mastery	-0.2647836850	-0.30640263
EI_Easy work	0.6363951076	0.29089569
EI_Pleasing	1.0000000000	0.24750942
EI_Dependence on teacher	0.2475094186	1.00000000

MC_Self-achievement	-0.2249210451	-0.06933198
MC_Power	0.0209222352	0.15547506
NE_Neuroticism	0.2143169837	0.06738510
NE_Openness	-0.1953551754	-0.05374545
NE_Conscientiousness	-0.1497863568	0.14006457
Emotional Intelligence	-0.1553007019	0.07097608
Letter Sets	0.0131172968	-0.11734590
Figure Classification	-0.1062366626	-0.15483213
GPA	-0.1435770263	0.02682367

	MC_Self-achievement	MC_Power	NE_Neuroticism
CN_Security	0.26511298	0.108645645	0.110982666
CN_Acceptance	0.23303799	0.076214871	0.099468536
CN_Initiative	0.41872634	0.188945449	-0.037930650
CN_Competence	0.45293048	0.242564828	0.005573149
CN_Meaning	0.37551762	0.163451827	0.003100202
M_Safety	0.19285961	0.260624175	0.182147915
M_Belonging	0.38427611	0.186641035	0.017374056
M_Self-esteem	0.47985900	0.303227304	0.002586548
M_Self-actualization	0.44546411	0.183841499	-0.120153881
EI_Challenge	0.58659615	0.067512875	-0.254197278
EI_Curiosity	0.49848352	0.134970538	-0.218196746
EI_Independence Mastery	0.37959811	0.029149436	-0.141426206
EI_Easy work	-0.42442019	0.059892169	0.310802579
EI_Pleasing	-0.22492105	0.020922235	0.214316984
EI_Dependence on teacher	-0.06933198	0.155475056	0.067385103
MC_Self-achievement	1.00000000	0.161899585	-0.154184612
MC_Power	0.16189959	1.000000000	0.044051668
NE_Neuroticism	-0.15418461	0.044051668	1.000000000
NE_Openness	0.20811755	0.008493264	0.042079358
NE_Conscientiousness	0.44595487	0.003326708	-0.321788622
Emotional Intelligence	0.50364356	0.176639099	-0.425319270
Letter Sets	0.07724483	-0.097694388	0.032933372
Figure Classification	-0.02666903	-0.113333367	0.052661754
GPA	0.08567961	0.065328635	-0.010905513

	NE_Openness	NE_Conscientiousness
CN_Security	0.110221983	0.230538721
CN_Acceptance	0.175860934	0.203111511
CN_Initiative	0.105738591	0.247265845
CN_Competence	0.049596804	0.283608212
CN_Meaning	0.078001782	0.296113136
M_Safety	-0.122620517	0.244471558
M_Belonging	0.110870788	0.303234716
M_Self-esteem	0.103350041	0.280445613

M_Self-actualization	0.261017218	0.186829814
EI_Challenge	0.217600146	0.325397041
EI_Curiosity	0.236302248	0.302367753
EI_Independence Mastery	0.059183217	0.261527149
EI_Easy work	-0.275266105	-0.240660464
EI_Pleasing	-0.195355175	-0.149786357
EI_Dependence on teacher	-0.053745455	0.140064574
MC_Self-achievement	0.208117548	0.445954868
MC_Power	0.008493264	0.003326708
NE_Neuroticism	0.042079358	-0.321788622
NE_Openness	1.000000000	-0.102340924
NE_Conscientiousness	-0.102340924	1.000000000
Emotional Intelligence	0.247773260	0.419333532
Letter Sets	0.062444116	0.009103993
Figure Classification	0.084112183	-0.063157691
GPA	0.006135441	0.205706091

	Emotional Intelligence	Letter Sets
CN_Security	0.28774582	-0.060368210
CN_Acceptance	0.28160730	-0.083451613
CN_Initiative	0.43373623	-0.124362289
CN_Competence	0.37936005	-0.061556236
CN_Meaning	0.38922262	-0.173030856
M_Safety	0.14966472	-0.122673444
M_Belonging	0.41307600	0.030652082
M_Self-esteem	0.45602674	-0.030881628
M_Self-actualization	0.49998446	-0.118005268
EI_Challenge	0.44228244	0.019303656
EI_Curiosity	0.51134336	-0.114909949
EI_Independence Mastery	0.25570133	0.064684360
EI_Easy work	-0.26395248	-0.073125909
EI_Pleasing	-0.15530070	0.013117297
EI_Dependence on teacher	0.07097608	-0.117345897
MC_Self-achievement	0.50364356	0.077244830
MC_Power	0.17663910	-0.097694388
NE_Neuroticism	-0.42531927	0.032933372
NE_Openness	0.24777326	0.062444116
NE_Conscientiousness	0.41933353	0.009103993
Emotional Intelligence	1.00000000	-0.112720570
Letter Sets	-0.11272057	1.000000000
Figure Classification	-0.12546414	0.456398124
GPA	0.07426018	0.063219463

	Figure Classification	GPA
CN_Security	-0.05474368	0.0004787008

```
CN_Acceptance                -0.05117797  0.0575669887
CN_Initiative                -0.21344052 -0.0924685729
CN_Competence                -0.11371674 -0.0263324090
CN_Meaning                   -0.16296513 -0.0727772843
M_Safety                     -0.15193046 -0.0813071199
M_Belonging                  -0.02276039  0.1023350503
M_Self-esteem                -0.09186538 -0.0292002029
M_Self-actualization         -0.13448426 -0.1789270226
EI_Challenge                  0.03202353  0.1107931879
EI_Curiosity                 -0.04457247  0.0725089358
EI_Independence Mastery       0.04283845  0.0994581347
EI_Easy work                 -0.14602276 -0.1590620210
EI_Pleasing                  -0.10623666 -0.1435770263
EI_Dependence on teacher     -0.15483213  0.0268236691
MC_Self-achievement          -0.02666903  0.0856796139
MC_Power                     -0.11333337  0.0653286351
NE_Neuroticism                0.05266175 -0.0109055134
NE_Openness                   0.08411218  0.0061354410
NE_Conscientiousness         -0.06315769  0.2057060914
Emotional Intelligence       -0.12546414  0.0742601785
Letter Sets                   0.45639812  0.0632194628
Figure Classification         1.00000000  0.1114047545
GPA                           0.11140475  1.0000000000
```

```
corr_test <- psych::corr.test(cor_data, use = "pairwise", method = "pearson",
                              adjust = "none")
print(corr_test$r)|>  kable (digits=2)   # Correlation coefficients
```

```
                          CN_Security CN_Acceptance CN_Initiative
CN_Security                1.0000000000   0.663484032    0.53300069
CN_Acceptance              0.6634840315   1.000000000    0.55038056
CN_Initiative              0.5330006854   0.550380559    1.00000000
CN_Competence              0.5768268480   0.488480266    0.69712508
CN_Meaning                 0.4821872957   0.467468495    0.61381630
M_Safety                   0.5090872217   0.321604258    0.35718742
M_Belonging                0.5656698827   0.448376915    0.36027607
M_Self-esteem              0.5932860068   0.461417046    0.58801900
M_Self-actualization       0.3792224152   0.425917831    0.58816997
EI_Challenge               0.0801164315   0.151013552    0.35226347
EI_Curiosity               0.2206242515   0.262404321    0.40317234
EI_Independence Mastery    0.1141094091   0.098354285    0.26279391
EI_Easy work               0.0996994300   0.001456188   -0.10349605
```

EI_Pleasing	0.0717706432	0.013437565	-0.11384500	
EI_Dependence on teacher	0.1976054689	0.183910357	0.08053243	
MC_Self-achievement	0.2651129756	0.233037992	0.41872634	
MC_Power	0.1086456450	0.076214871	0.18894545	
NE_Neuroticism	0.1109826662	0.099468536	-0.03793065	
NE_Openness	0.1102219831	0.175860934	0.10573859	
NE_Conscientiousness	0.2305387208	0.203111511	0.24726585	
Emotional Intelligence	0.2877458214	0.281607300	0.43373623	
Letter Sets	-0.0603682099	-0.083451613	-0.12436229	
Figure Classification	-0.0547436816	-0.051177969	-0.21344052	
GPA	0.0004787008	0.057566989	-0.09246857	
	CN_Competence	CN_Meaning	M_Safety	M_Belonging
CN_Security	0.5768268480	0.482187296	0.50908722	0.56566988
CN_Acceptance	0.4884802657	0.467468495	0.32160426	0.44837691
CN_Initiative	0.6971250820	0.613816296	0.35718742	0.36027607
CN_Competence	1.0000000000	0.597788058	0.44295511	0.41611965
CN_Meaning	0.5977880581	1.000000000	0.41816707	0.41255781
M_Safety	0.4429551135	0.418167073	1.00000000	0.39903988
M_Belonging	0.4161196545	0.412557807	0.39903988	1.00000000
M_Self-esteem	0.7198862878	0.581344865	0.52708542	0.61405307
M_Self-actualization	0.4855572253	0.482808618	0.30641109	0.42277839
EI_Challenge	0.2692522559	0.322613268	0.03345958	0.14546967
EI_Curiosity	0.3209193826	0.416982531	0.15493423	0.25179307
EI_Independence Mastery	0.2414139815	0.231000417	0.15297408	0.07820184
EI_Easy work	-0.0421848180	-0.029239573	0.21671306	0.04683165
EI_Pleasing	0.0007547254	-0.042975025	0.15315228	0.05250666
EI_Dependence on teacher	0.1080213021	0.165443073	0.18508170	0.21699956
MC_Self-achievement	0.4529304801	0.375517618	0.19285961	0.38427611
MC_Power	0.2425648281	0.163451827	0.26062418	0.18664104
NE_Neuroticism	0.0055731489	0.003100202	0.18214791	0.01737406
NE_Openness	0.0495968040	0.078001782	-0.12262052	0.11087079
NE_Conscientiousness	0.2836082121	0.296113136	0.24447156	0.30323472
Emotional Intelligence	0.3793600460	0.389222624	0.14966472	0.41307600
Letter Sets	-0.0615562359	-0.173030856	-0.12267344	0.03065208
Figure Classification	-0.1137167377	-0.162965133	-0.15193046	-0.02276039
GPA	-0.0263324090	-0.072777284	-0.08130712	0.10233505
	M_Self-esteem	M_Self-actualization	EI_Challenge	
CN_Security	0.593286007	0.37922242	0.08011643	
CN_Acceptance	0.461417046	0.42591783	0.15101355	
CN_Initiative	0.588018996	0.58816997	0.35226347	
CN_Competence	0.719886288	0.48555723	0.26925226	
CN_Meaning	0.581344865	0.48280862	0.32261327	
M_Safety	0.527085423	0.30641109	0.03345958	

M_Belonging	0.614053070	0.42277839	0.14546967
M_Self-esteem	1.000000000	0.61817064	0.23024485
M_Self-actualization	0.618170642	1.00000000	0.27352684
EI_Challenge	0.230244852	0.27352684	1.00000000
EI_Curiosity	0.306148766	0.36352734	0.75882685
EI_Independence Mastery	0.167364821	0.13329756	0.59276670
EI_Easy work	0.033073260	-0.05740910	-0.69140791
EI_Pleasing	0.044558892	-0.03398211	-0.45313859
EI_Dependence on teacher	0.162796918	0.08897508	-0.14490814
MC_Self-achievement	0.479859003	0.44546411	0.58659615
MC_Power	0.303227304	0.18384150	0.06751288
NE_Neuroticism	0.002586548	-0.12015388	-0.25419728
NE_Openness	0.103350041	0.26101722	0.21760015
NE_Conscientiousness	0.280445613	0.18682981	0.32539704
Emotional Intelligence	0.456026741	0.49998446	0.44228244
Letter Sets	-0.030881628	-0.11800527	0.01930366
Figure Classification	-0.091865379	-0.13448426	0.03202353
GPA	-0.029200203	-0.17892702	0.11079319

	EI_Curiosity	EI_Independence Mastery	EI_Easy work
CN_Security	0.22062425	0.11410941	0.099699430
CN_Acceptance	0.26240432	0.09835429	0.001456188
CN_Initiative	0.40317234	0.26279391	-0.103496054
CN_Competence	0.32091938	0.24141398	-0.042184818
CN_Meaning	0.41698253	0.23100042	-0.029239573
M_Safety	0.15493423	0.15297408	0.216713057
M_Belonging	0.25179307	0.07820184	0.046831652
M_Self-esteem	0.30614877	0.16736482	0.033073260
M_Self-actualization	0.36352734	0.13329756	-0.057409099
EI_Challenge	0.75882685	0.59276670	-0.691407911
EI_Curiosity	1.00000000	0.47309479	-0.507426395
EI_Independence Mastery	0.47309479	1.00000000	-0.405823855
EI_Easy work	-0.50742640	-0.40582386	1.000000000
EI_Pleasing	-0.43054442	-0.26478368	0.636395108
EI_Dependence on teacher	0.05674784	-0.30640263	0.290895693
MC_Self-achievement	0.49848352	0.37959811	-0.424420189
MC_Power	0.13497054	0.02914944	0.059892169
NE_Neuroticism	-0.21819675	-0.14142621	0.310802579
NE_Openness	0.23630225	0.05918322	-0.275266105
NE_Conscientiousness	0.30236775	0.26152715	-0.240660464
Emotional Intelligence	0.51134336	0.25570133	-0.263952479
Letter Sets	-0.11490995	0.06468436	-0.073125909
Figure Classification	-0.04457247	0.04283845	-0.146022765
GPA	0.07250894	0.09945813	-0.159062021

	EI_Pleasing	EI_Dependence on teacher
CN_Security	0.0717706432	0.19760547
CN_Acceptance	0.0134375652	0.18391036
CN_Initiative	-0.1138449977	0.08053243
CN_Competence	0.0007547254	0.10802130
CN_Meaning	-0.0429750246	0.16544307
M_Safety	0.1531522816	0.18508170
M_Belonging	0.0525066597	0.21699956
M_Self-esteem	0.0445588919	0.16279692
M_Self-actualization	-0.0339821080	0.08897508
EI_Challenge	-0.4531385858	-0.14490814
EI_Curiosity	-0.4305444184	0.05674784
EI_Independence Mastery	-0.2647836850	-0.30640263
EI_Easy work	0.6363951076	0.29089569
EI_Pleasing	1.0000000000	0.24750942
EI_Dependence on teacher	0.2475094186	1.00000000
MC_Self-achievement	-0.2249210451	-0.06933198
MC_Power	0.0209222352	0.15547506
NE_Neuroticism	0.2143169837	0.06738510
NE_Openness	-0.1953551754	-0.05374545
NE_Conscientiousness	-0.1497863568	0.14006457
Emotional Intelligence	-0.1553007019	0.07097608
Letter Sets	0.0131172968	-0.11734590
Figure Classification	-0.1062366626	-0.15483213
GPA	-0.1435770263	0.02682367

	MC_Self-achievement	MC_Power	NE_Neuroticism
CN_Security	0.26511298	0.108645645	0.110982666
CN_Acceptance	0.23303799	0.076214871	0.099468536
CN_Initiative	0.41872634	0.188945449	-0.037930650
CN_Competence	0.45293048	0.242564828	0.005573149
CN_Meaning	0.37551762	0.163451827	0.003100202
M_Safety	0.19285961	0.260624175	0.182147915
M_Belonging	0.38427611	0.186641035	0.017374056
M_Self-esteem	0.47985900	0.303227304	0.002586548
M_Self-actualization	0.44546411	0.183841499	-0.120153881
EI_Challenge	0.58659615	0.067512875	-0.254197278
EI_Curiosity	0.49848352	0.134970538	-0.218196746
EI_Independence Mastery	0.37959811	0.029149436	-0.141426206
EI_Easy work	-0.42442019	0.059892169	0.310802579
EI_Pleasing	-0.22492105	0.020922235	0.214316984
EI_Dependence on teacher	-0.06933198	0.155475056	0.067385103
MC_Self-achievement	1.00000000	0.161899585	-0.154184612
MC_Power	0.16189959	1.000000000	0.044051668

NE_Neuroticism	-0.15418461	0.044051668	1.000000000
NE_Openness	0.20811755	0.008493264	0.042079358
NE_Conscientiousness	0.44595487	0.003326708	-0.321788622
Emotional Intelligence	0.50364356	0.176639099	-0.425319270
Letter Sets	0.07724483	-0.097694388	0.032933372
Figure Classification	-0.02666903	-0.113333367	0.052661754
GPA	0.08567961	0.065328635	-0.010905513

	NE_Openness	NE_Conscientiousness
CN_Security	0.110221983	0.230538721
CN_Acceptance	0.175860934	0.203111511
CN_Initiative	0.105738591	0.247265845
CN_Competence	0.049596804	0.283608212
CN_Meaning	0.078001782	0.296113136
M_Safety	-0.122620517	0.244471558
M_Belonging	0.110870788	0.303234716
M_Self-esteem	0.103350041	0.280445613
M_Self-actualization	0.261017218	0.186829814
EI_Challenge	0.217600146	0.325397041
EI_Curiosity	0.236302248	0.302367753
EI_Independence Mastery	0.059183217	0.261527149
EI_Easy work	-0.275266105	-0.240660464
EI_Pleasing	-0.195355175	-0.149786357
EI_Dependence on teacher	-0.053745455	0.140064574
MC_Self-achievement	0.208117548	0.445954868
MC_Power	0.008493264	0.003326708
NE_Neuroticism	0.042079358	-0.321788622
NE_Openness	1.000000000	-0.102340924
NE_Conscientiousness	-0.102340924	1.000000000
Emotional Intelligence	0.247773260	0.419333532
Letter Sets	0.062444116	0.009103993
Figure Classification	0.084112183	-0.063157691
GPA	0.006135441	0.205706091

	Emotional Intelligence	Letter Sets
CN_Security	0.28774582	-0.060368210
CN_Acceptance	0.28160730	-0.083451613
CN_Initiative	0.43373623	-0.124362289
CN_Competence	0.37936005	-0.061556236
CN_Meaning	0.38922262	-0.173030856
M_Safety	0.14966472	-0.122673444
M_Belonging	0.41307600	0.030652082
M_Self-esteem	0.45602674	-0.030881628
M_Self-actualization	0.49998446	-0.118005268
EI_Challenge	0.44228244	0.019303656

EI_Curiosity	0.51134336	-0.114909949
EI_Independence Mastery	0.25570133	0.064684360
EI_Easy work	-0.26395248	-0.073125909
EI_Pleasing	-0.15530070	0.013117297
EI_Dependence on teacher	0.07097608	-0.117345897
MC_Self-achievement	0.50364356	0.077244830
MC_Power	0.17663910	-0.097694388
NE_Neuroticism	-0.42531927	0.032933372
NE_Openness	0.24777326	0.062444116
NE_Conscientiousness	0.41933353	0.009103993
Emotional Intelligence	1.00000000	-0.112720570
Letter Sets	-0.11272057	1.000000000
Figure Classification	-0.12546414	0.456398124
GPA	0.07426018	0.063219463

	Figure Classification	GPA
CN_Security	-0.05474368	0.0004787008
CN_Acceptance	-0.05117797	0.0575669887
CN_Initiative	-0.21344052	-0.0924685729
CN_Competence	-0.11371674	-0.0263324090
CN_Meaning	-0.16296513	-0.0727772843
M_Safety	-0.15193046	-0.0813071199
M_Belonging	-0.02276039	0.1023350503
M_Self-esteem	-0.09186538	-0.0292002029
M_Self-actualization	-0.13448426	-0.1789270226
EI_Challenge	0.03202353	0.1107931879
EI_Curiosity	-0.04457247	0.0725089358
EI_Independence Mastery	0.04283845	0.0994581347
EI_Easy work	-0.14602276	-0.1590620210
EI_Pleasing	-0.10623666	-0.1435770263
EI_Dependence on teacher	-0.15483213	0.0268236691
MC_Self-achievement	-0.02666903	0.0856796139
MC_Power	-0.11333337	0.0653286351
NE_Neuroticism	0.05266175	-0.0109055134
NE_Openness	0.08411218	0.0061354410
NE_Conscientiousness	-0.06315769	0.2057060914
Emotional Intelligence	-0.12546414	0.0742601785
Letter Sets	0.45639812	0.0632194628
Figure Classification	1.00000000	0.1114047545
GPA	0.11140475	1.0000000000

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence	CN_Meaningful	M_Safety	M_Belonging	M_Esteem	M_Self-actualization	EI_Challenge	EI_Curiosity	EI_Independence	EI_Mastery	EI_Dependence	EI_Easy	MC_Pleasant	MC_Achievement	NE_Self-criticism	NE_Nervousness	NE_Openness	NE_Emotional	NE_Concentration	NE_Intelligences	Letter Sets	Figure Classification	GPA
CN_Security	1.00	0.66	0.53	0.58	0.48	0.51	0.57	0.59	0.38	0.08	0.22	0.11	0.10	0.07	0.20	0.27	0.11	0.11	0.11	0.23	0.29	-0.06	-0.05	0.00
CN_Acceptance	0.66	1.00	0.55	0.49	0.47	0.32	0.45	0.46	0.43	0.15	0.26	0.10	0.00	0.01	0.18	0.23	0.08	0.10	0.18	0.20	0.28	-0.08	-0.05	0.06
CN_Initiative	0.53	0.55	1.00	0.70	0.61	0.36	0.36	0.59	0.59	0.35	0.40	0.26	-0.10	-0.11	0.08	0.42	0.19	-0.04	0.11	0.25	0.43	-0.12	-0.21	-0.09
CN_Competence	0.58	0.49	0.70	1.00	0.60	0.44	0.42	0.72	0.49	0.27	0.32	0.24	-0.04	0.00	0.11	0.45	0.24	0.01	0.05	0.28	0.38	-0.06	-0.11	-0.03
CN_Meaningful	0.48	0.47	0.61	0.60	1.00	0.42	0.41	0.58	0.48	0.32	0.42	0.23	-0.03	-0.04	0.17	0.38	0.16	0.00	0.08	0.30	0.39	-0.17	-0.16	-0.07
M_Safety	0.51	0.32	0.36	0.44	0.42	1.00	0.40	0.53	0.31	0.03	0.15	0.15	0.22	0.15	0.19	0.19	0.26	0.18	-0.12	0.24	0.15	-0.12	-0.15	-0.08
M_Belonging	0.57	0.45	0.36	0.42	0.41	0.40	1.00	0.61	0.42	0.15	0.25	0.08	0.05	0.05	0.22	0.38	0.19	0.02	0.11	0.30	0.41	0.03	-0.02	-0.10
M_Esteem	0.59	0.46	0.59	0.72	0.58	0.53	0.61	1.00	0.62	0.23	0.31	0.17	0.03	0.04	0.16	0.48	0.30	0.00	0.10	0.28	0.46	-0.03	-0.09	-0.03
M_Self-actualization	0.38	0.43	0.59	0.49	0.48	0.31	0.42	0.62	1.00	0.27	0.36	0.13	-0.06	-0.03	0.09	0.45	0.18	-0.12	0.26	0.19	0.50	-0.12	-0.13	-0.18
EI_Challenge	0.08	0.15	0.35	0.27	0.32	0.03	0.15	0.23	0.27	1.00	0.76	0.59	-0.69	-0.45	-0.14	0.59	0.07	-0.25	0.22	0.33	0.44	0.02	0.03	0.11
EI_Curiosity	0.22	0.26	0.40	0.32	0.42	0.15	0.25	0.31	0.36	0.76	1.00	0.47	-0.51	-0.43	0.06	0.50	0.13	-0.22	0.24	0.30	0.51	-0.11	-0.04	0.07
EI_Independence	0.11	0.10	0.26	0.24	0.23	0.15	0.08	0.17	0.13	0.59	0.47	1.00	-0.41	-0.26	-0.31	0.38	0.03	-0.14	0.06	0.26	0.26	0.06	0.04	0.10
EI_Dependence																								
Mastery																								

	CN_Curiosity	CN_Self-acceptance	CN_Achievement	CN_Initiative	CN_Common Mission	CN_Meaning in life	CN_Safety	CN_Self-actualization	CN_Self-regulation	CN_Classroom Mastery	EI_Independence	EI_Dependence	EI_Easy going	EI_Pleasing	MC_Self-achievement	MC_Motivation	NE_Neuroticism	NE_Openness	NE_Emotional Intelligence	NE_Conscientiousness	Letter Sets	Figure Classification	GPA
EI_Easy work	0.10	0.00	-0.10	-0.04	-0.03	0.22	0.05	0.03	-0.06	-0.69	-0.51	-0.41	1.00	0.64	0.29	-0.42	0.06	0.31	-0.28	-0.24	-0.26	0.07	0.15	0.16
EI_Pleasing	-0.01	-0.11	0.00	-0.04	0.15	0.05	0.04	-0.03	-0.45	-0.43	-0.26	0.64	1.00	0.25	-0.22	0.02	0.21	-0.20	-0.15	-0.16	0.01	-0.11	0.14
EI_Dependence on teacher	0.20	0.18	0.08	0.11	0.17	0.19	0.22	0.16	0.09	-0.14	0.06	-0.31	0.29	0.25	1.00	-0.07	0.16	0.07	-0.05	0.14	0.07	-0.12	-0.15	0.03
MC_Self-achievement	0.27	0.23	0.42	0.45	0.38	0.19	0.38	0.48	0.45	0.59	0.50	0.38	-0.42	-0.22	-0.07	1.00	0.16	-0.15	0.21	0.45	0.50	0.08	-0.03	0.09
MC_Motivation	0.16	0.08	0.19	0.24	0.16	0.26	0.19	0.30	0.18	0.07	0.13	0.03	0.06	0.02	0.16	0.16	1.00	0.04	0.01	0.00	0.18	-0.10	-0.11	0.07
NE_Neuroticism	-0.10	-0.04	0.01	0.00	0.18	0.02	0.00	-0.12	-0.25	-0.22	-0.14	0.31	0.21	0.07	-0.15	0.04	1.00	0.04	-0.32	-0.43	0.03	0.05	-0.01
NE_Openness	0.18	0.11	0.05	0.08	-0.12	0.11	0.10	0.26	0.22	0.24	0.06	-0.28	-0.20	-0.05	0.21	0.01	0.04	1.00	-0.10	0.25	0.06	0.08	0.01
NE_Conscientiousness	0.23	0.20	0.25	0.28	0.30	0.24	0.30	0.28	0.19	0.33	0.30	0.26	-0.24	-0.15	0.14	0.45	0.00	-0.32	-0.10	1.00	0.42	0.01	-0.06	0.21
Emotional Intelligence	0.29	0.28	0.43	0.38	0.39	0.15	0.41	0.46	0.50	0.44	0.51	0.26	-0.26	-0.16	0.07	0.50	0.18	-0.43	0.25	0.42	1.00	-0.11	-0.13	0.07
Letter Sets	-0.06	-0.08	-0.12	-0.06	-0.17	-0.12	0.03	-0.03	-0.12	0.02	-0.11	0.06	-0.07	0.01	-0.12	0.08	-0.10	0.03	0.06	0.01	-0.11	1.00	0.46	0.06

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence	CN_Meaning	M_Safety	M_Belonging	M_Self-esteem	M_Self-actualization	EI_Challenge	EI_Curiosity	EI_Independence	EI_Dependence	EI_Pleasantness	MC_Self-actualization	MC_Curiosity	NE_Neuroticism	NE_Openness	Emotional Intelligence	Communication	Letters	Figure Classification	GPA
Figure Classification	-0.05	-0.05	-0.21	-0.11	-0.16	-0.15	-0.02	-0.09	-0.13	0.03	0.04	-0.15	-0.11	-0.15	-0.03	-0.11	0.05	0.08	0.06	-0.13	0.46	1.00	0.11
GPA	0.00	0.06	-0.09	-0.03	-0.07	-0.08	0.10	-0.03	-0.18	0.11	0.07	0.10	-0.16	-0.14	0.03	0.09	0.07	-0.01	0.01	0.21	0.07	0.06	0.11	1.00

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print(corr_test$p)|>  kable (digits=2)    # p-values
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	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence
CN_Security	0.000000e+00	7.956571e-51	5.251954e-30	5.735923e-36
CN_Acceptance	7.956571e-51	0.000000e+00	2.890906e-32	8.838440e-25
CN_Initiative	5.251954e-30	2.890906e-32	0.000000e+00	4.924170e-58
CN_Competence	5.735923e-36	8.838440e-25	4.924170e-58	0.000000e+00
CN_Meaning	4.218226e-24	1.441885e-22	9.530217e-42	3.764424e-39
M_Safety	4.195161e-27	7.801063e-11	3.528039e-13	3.582455e-20
M_Belonging	2.297490e-34	1.098864e-20	2.136247e-13	9.185018e-18
M_Self-esteem	1.901200e-38	5.865710e-22	1.224476e-37	1.654883e-63
M_Self-actualization	8.721673e-15	1.283307e-18	1.161368e-37	1.833998e-24
EI_Challenge	1.141906e-01	2.790957e-03	7.762927e-13	6.653344e-08
EI_Curiosity	1.096078e-05	1.457471e-07	1.123117e-16	8.597662e-11

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EI_Independence Mastery  2.422055e-02  5.228074e-02  1.394700e-07  1.408677e-
06
EI_Easy work             4.912668e-02  9.771317e-01  4.107111e-02  4.060984e-
01
EI_Pleasing              1.571764e-01  7.913709e-01  2.455348e-02  9.881464e-
01
EI_Dependence on teacher 8.541507e-05  2.607401e-04  1.123173e-01  3.295376e-
02
MC_Self-achievement      1.071607e-07  3.294393e-06  5.475254e-18  4.006327e-
21
MC_Power                 3.194918e-02  1.329737e-01  1.745514e-04  1.250407e-
06
NE_Neuroticism           2.841773e-02  4.965648e-02  4.551030e-01  9.126406e-
01
NE_Openness              2.952851e-02  4.847273e-04  3.685905e-02  3.286108e-
01
NE_Conscientiousness     4.219358e-06  5.334895e-05  7.637040e-07  1.195996e-
08
Emotional Intelligence   7.159933e-09  1.528227e-08  2.546930e-19  8.514770e-
15
Letter Sets              2.366982e-01  1.016089e-01  1.448901e-02  2.275822e-
01
Figure Classification    2.814591e-01  3.140314e-01  2.184218e-05  2.490281e-
02
GPA                      9.928392e-01  2.800613e-01  8.232427e-02  6.214653e-
01
                           CN_Meaning     M_Safety  M_Belonging M_Self-esteem
CN_Security              4.218226e-24 4.195161e-27 2.297490e-34  1.901200e-38
CN_Acceptance            1.441885e-22 7.801063e-11 1.098864e-20  5.865710e-22
CN_Initiative            9.530217e-42 3.528039e-13 2.136247e-13  1.224476e-37
CN_Competence            3.764424e-39 3.582455e-20 9.185018e-18  1.654883e-63
CN_Meaning               0.000000e+00 6.120313e-18 1.848991e-17  1.235836e-36
M_Safety                 6.120313e-18 0.000000e+00 2.441343e-16  2.883005e-29
M_Belonging              1.848991e-17 2.441343e-16 0.000000e+00  8.702128e-42
M_Self-esteem            1.235836e-36 2.883005e-29 8.702128e-42  0.000000e+00
M_Self-actualization     3.620241e-24 6.369320e-10 2.427985e-18  1.768963e-42
EI_Challenge             6.756904e-11 5.100003e-01 3.990606e-03  4.343145e-06
EI_Curiosity             7.743251e-18 2.151843e-03 4.705416e-07  6.597613e-10
EI_Independence Mastery  4.031654e-06 2.452456e-03 1.231302e-01  9.065990e-04
EI_Easy work             5.648146e-01 1.578655e-05 3.563259e-01  5.149005e-01
EI_Pleasing              3.973544e-01 2.423623e-03 3.009919e-01  3.801722e-01
EI_Dependence on teacher 1.040360e-03 2.377202e-04 1.537369e-05  1.254413e-03
MC_Self-achievement      1.657400e-14 1.268547e-04 3.584985e-15  7.459690e-24
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MC_Power	1.197959e-03	1.780613e-07	2.100144e-04	9.742908e-10
NE_Neuroticism	9.513371e-01	2.993283e-04	7.323249e-01	9.593922e-01
NE_Openness	1.240948e-01	1.539554e-02	2.857882e-02	4.135890e-02
NE_Conscientiousness	2.472655e-09	1.025002e-06	9.733330e-10	1.760392e-08
Emotional Intelligence	1.479268e-15	3.047959e-03	1.670912e-17	1.999909e-21
Letter Sets	6.395089e-04	1.588929e-02	5.482358e-01	5.452426e-01
Figure Classification	1.257884e-03	2.661463e-03	6.545008e-01	7.031651e-02
GPA	1.718555e-01	1.267852e-01	5.439651e-02	5.839857e-01

	M_Self-actualization	EI_Challenge	EI_Curiosity
CN_Security	8.721673e-15	1.141906e-01	1.096078e-05
CN_Acceptance	1.283307e-18	2.790957e-03	1.457471e-07
CN_Initiative	1.161368e-37	7.762927e-13	1.123117e-16
CN_Competence	1.833998e-24	6.653344e-08	8.597662e-11
CN_Meaning	3.620241e-24	6.756904e-11	7.743251e-18
M_Safety	6.369320e-10	5.100003e-01	2.151843e-03
M_Belonging	2.427985e-18	3.990606e-03	4.705416e-07
M_Self-esteem	1.768963e-42	4.343145e-06	6.597613e-10
M_Self-actualization	0.000000e+00	4.032643e-08	1.252383e-13
EI_Challenge	4.032643e-08	0.000000e+00	2.960615e-74
EI_Curiosity	1.252383e-13	2.960615e-74	0.000000e+00
EI_Independence Mastery	8.395448e-03	2.287944e-38	3.813944e-23
EI_Easy work	2.580413e-01	9.708070e-57	6.545213e-27
EI_Pleasing	5.034113e-01	3.824392e-21	4.953768e-19
EI_Dependence on teacher	7.926631e-02	4.135048e-03	2.635739e-01
MC_Self-achievement	2.078819e-20	2.013467e-37	6.888135e-26
MC_Power	2.621561e-04	1.833499e-01	7.604970e-03
NE_Neuroticism	1.760425e-02	3.624507e-07	1.375687e-05
NE_Openness	1.703815e-07	1.454126e-05	2.374650e-06
NE_Conscientiousness	2.068732e-04	4.532824e-11	1.091802e-09
Emotional Intelligence	4.662723e-26	4.142080e-20	2.283648e-27
Letter Sets	2.039370e-02	7.053846e-01	2.395913e-02
Figure Classification	7.907739e-03	5.288701e-01	3.806425e-01
GPA	7.198197e-04	3.719590e-02	1.734458e-01

	EI_Independence Mastery	EI_Easy work	EI_Pleasing
CN_Security	2.422055e-02	4.912668e-02	1.571764e-01
CN_Acceptance	5.228074e-02	9.771317e-01	7.913709e-01
CN_Initiative	1.394700e-07	4.107111e-02	2.455348e-02
CN_Competence	1.408677e-06	4.060984e-01	9.881464e-01
CN_Meaning	4.031654e-06	5.648146e-01	3.973544e-01
M_Safety	2.452456e-03	1.578655e-05	2.423623e-03
M_Belonging	1.231302e-01	3.563259e-01	3.009919e-01
M_Self-esteem	9.065990e-04	5.149005e-01	3.801722e-01
M_Self-actualization	8.395448e-03	2.580413e-01	5.034113e-01

EI_Challenge	2.287944e-38	9.708070e-57	3.824392e-21
EI_Curiosity	3.813944e-23	6.545213e-27	4.953768e-19
EI_Independence Mastery	0.000000e+00	6.785440e-17	1.112635e-07
EI_Easy work	6.785440e-17	0.000000e+00	1.143845e-45
EI_Pleasing	1.112635e-07	1.143845e-45	0.000000e+00
EI_Dependence on teacher	6.376561e-10	4.817746e-09	7.442386e-07
MC_Self-achievement	8.168185e-15	1.740972e-18	7.285393e-06
MC_Power	5.660156e-01	2.379849e-01	6.804128e-01
NE_Neuroticism	5.141127e-03	3.514019e-10	1.967609e-05
NE_Openness	2.435960e-01	3.281201e-08	1.031548e-04
NE_Conscientiousness	1.608920e-07	1.522514e-06	3.023931e-03
Emotional Intelligence	3.074344e-07	1.223045e-07	2.099524e-03
Letter Sets	2.047791e-01	1.515845e-01	7.972643e-01
Figure Classification	3.994667e-01	3.898791e-03	3.621527e-02
GPA	6.157935e-02	2.688477e-03	6.812687e-03

	EI_Dependence on teacher	MC_Self-achievement
CN_Security	8.541507e-05	1.071607e-07
CN_Acceptance	2.607401e-04	3.294393e-06
CN_Initiative	1.123173e-01	5.475254e-18
CN_Competence	3.295376e-02	4.006327e-21
CN_Meaning	1.040360e-03	1.657400e-14
M_Safety	2.377202e-04	1.268547e-04
M_Belonging	1.537369e-05	3.584985e-15
M_Self-esteem	1.254413e-03	7.459690e-24
M_Self-actualization	7.926631e-02	2.078819e-20
EI_Challenge	4.135048e-03	2.013467e-37
EI_Curiosity	2.635739e-01	6.888135e-26
EI_Independence Mastery	6.376561e-10	8.168185e-15
EI_Easy work	4.817746e-09	1.740972e-18
EI_Pleasing	7.442386e-07	7.285393e-06
EI_Dependence on teacher	0.000000e+00	1.717988e-01
MC_Self-achievement	1.717988e-01	0.000000e+00
MC_Power	2.075042e-03	1.335740e-03
NE_Neuroticism	1.841822e-01	2.262590e-03
NE_Openness	2.897154e-01	3.439260e-05
NE_Conscientiousness	5.591082e-03	1.867903e-20
Emotional Intelligence	1.618337e-01	1.786422e-26
Letter Sets	2.111196e-02	1.297819e-01
Figure Classification	2.195498e-03	6.000031e-01
GPA	6.149714e-01	1.075483e-01

	MC_Power	NE_Neuroticism	NE_Openness
CN_Security	3.194918e-02	2.841773e-02	2.952851e-02
CN_Acceptance	1.329737e-01	4.965648e-02	4.847273e-04

CN_Initiative	1.745514e-04	4.551030e-01	3.685905e-02
CN_Competence	1.250407e-06	9.126406e-01	3.286108e-01
CN_Meaning	1.197959e-03	9.513371e-01	1.240948e-01
M_Safety	1.780613e-07	2.993283e-04	1.539554e-02
M_Belonging	2.100144e-04	7.323249e-01	2.857882e-02
M_Self-esteem	9.742908e-10	9.593922e-01	4.135890e-02
M_Self-actualization	2.621561e-04	1.760425e-02	1.703815e-07
EI_Challenge	1.833499e-01	3.624507e-07	1.454126e-05
EI_Curiosity	7.604970e-03	1.375687e-05	2.374650e-06
EI_Independence Mastery	5.660156e-01	5.141127e-03	2.435960e-01
EI_Easy work	2.379849e-01	3.514019e-10	3.281201e-08
EI_Pleasing	6.804128e-01	1.967609e-05	1.031548e-04
EI_Dependence on teacher	2.075042e-03	1.841822e-01	2.897154e-01
MC_Self-achievement	1.335740e-03	2.262590e-03	3.439260e-05
MC_Power	0.000000e+00	3.856246e-01	8.672181e-01
NE_Neuroticism	3.856246e-01	0.000000e+00	4.072739e-01
NE_Openness	8.672181e-01	4.072739e-01	0.000000e+00
NE_Conscientiousness	9.477866e-01	7.599209e-11	4.339477e-02
Emotional Intelligence	4.570487e-04	1.449941e-18	7.236909e-07
Letter Sets	5.514209e-02	5.188529e-01	2.209340e-01
Figure Classification	2.539790e-02	3.001888e-01	9.761059e-02
GPA	2.201559e-01	8.379891e-01	9.084210e-01

	NE_Conscientiousness	Emotional Intelligence
CN_Security	4.219358e-06	7.159933e-09
CN_Acceptance	5.334895e-05	1.528227e-08
CN_Initiative	7.637040e-07	2.546930e-19
CN_Competence	1.195996e-08	8.514770e-15
CN_Meaning	2.472655e-09	1.479268e-15
M_Safety	1.025002e-06	3.047959e-03
M_Belonging	9.733330e-10	1.670912e-17
M_Self-esteem	1.760392e-08	1.999909e-21
M_Self-actualization	2.068732e-04	4.662723e-26
EI_Challenge	4.532824e-11	4.142080e-20
EI_Curiosity	1.091802e-09	2.283648e-27
EI_Independence Mastery	1.608920e-07	3.074344e-07
EI_Easy work	1.522514e-06	1.223045e-07
EI_Pleasing	3.023931e-03	2.099524e-03
EI_Dependence on teacher	5.591082e-03	1.618337e-01
MC_Self-achievement	1.867903e-20	1.786422e-26
MC_Power	9.477866e-01	4.570487e-04
NE_Neuroticism	7.599209e-11	1.449941e-18
NE_Openness	4.339477e-02	7.236909e-07
NE_Conscientiousness	0.000000e+00	4.850514e-18

Emotional Intelligence	4.850514e-18	0.000000e+00
Letter Sets	8.584963e-01	2.679526e-02
Figure Classification	2.139059e-01	1.327264e-02
GPA	9.682260e-05	1.632630e-01

	Letter Sets	Figure Classification	GPA
CN_Security	2.366982e-01	2.814591e-01	0.9928392040
CN_Acceptance	1.016089e-01	3.140314e-01	0.2800613060
CN_Initiative	1.448901e-02	2.184218e-05	0.0823242691
CN_Competence	2.275822e-01	2.490281e-02	0.6214653295
CN_Meaning	6.395089e-04	1.257884e-03	0.1718555450
M_Safety	1.588929e-02	2.661463e-03	0.1267851922
M_Belonging	5.482358e-01	6.545008e-01	0.0543965070
M_Self-esteem	5.452426e-01	7.031651e-02	0.5839856883
M_Self-actualization	2.039370e-02	7.907739e-03	0.0007198197
EI_Challenge	7.053846e-01	5.288701e-01	0.0371958966
EI_Curiosity	2.395913e-02	3.806425e-01	0.1734458440
EI_Independence Mastery	2.047791e-01	3.994667e-01	0.0615793532
EI_Easy work	1.515845e-01	3.898791e-03	0.0026884769
EI_Pleasing	7.972643e-01	3.621527e-02	0.0068126874
EI_Dependence on teacher	2.111196e-02	2.195498e-03	0.6149714025
MC_Self-achievement	1.297819e-01	6.000031e-01	0.1075482884
MC_Power	5.514209e-02	2.539790e-02	0.2201558546
NE_Neuroticism	5.188529e-01	3.001888e-01	0.8379890582
NE_Openness	2.209340e-01	9.761059e-02	0.9084209992
NE_Conscientiousness	8.584963e-01	2.139059e-01	0.0000968226
Emotional Intelligence	2.679526e-02	1.327264e-02	0.1632629801
Letter Sets	0.000000e+00	2.949134e-21	0.2381286150
Figure Classification	2.949134e-21	0.000000e+00	0.0364207954
GPA	2.381286e-01	3.642080e-02	0.0000000000

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence	CN_Meaning	M_Safety	M_Belonging	M_Self-esteem	M_Self-actualization	EI_Challenge	EI_Curiosity	EI_Independence Mastery	EI_Easy work	EI_Pleasing	EI_Dependence on teacher	MC_Self-achievement	MC_Power	NE_Neuroticism	NE_Openness	NE_Conscientiousness	Emotional Intelligence	Letter Sets	Figure Classification	GPA
CN_Security	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.02	0.05	0.16	0.00	0.00	0.03	0.03	0.03	0.00	0.00	0.24	0.28	0.99

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence	CN_Meaning	M_Belonging	M_Safety	M_Self-esteem	M_Self-actualization	EI_Challenge	EI_Curiosity	EI_Independence	EI_Mastery	EI_Easy	EI_Pleasing	MC_Self-actualization	MC_Meaningfulness	NE_Neuroticism	NE_Openness	Emotional Intelligence	Conscientiousness	Letter Sets	Figure Classification	GPA
CN_Acceptance	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.05	0.98	0.79	0.00	0.00	0.13	0.05	0.00	0.00	0.00	0.10	0.31	0.28
CN_Initiative	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.02	0.11	0.00	0.00	0.46	0.04	0.00	0.00	0.01	0.00	0.08
CN_Competence	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.41	0.99	0.03	0.00	0.00	0.91	0.33	0.00	0.00	0.23	0.02	0.62
CN_Meaning	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.56	0.40	0.00	0.00	0.00	0.95	0.12	0.00	0.00	0.00	0.00	0.17
M_Safety	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.02	0.00	0.13
M_Belonging	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.12	0.36	0.30	0.00	0.00	0.00	0.73	0.03	0.00	0.00	0.55	0.65	0.05
M_Self-esteem	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.51	0.38	0.00	0.00	0.00	0.96	0.04	0.00	0.00	0.55	0.07	0.58
M_Self-actualization	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.26	0.50	0.08	0.00	0.00	0.02	0.00	0.00	0.00	0.02	0.01	0.00
EI_Challenge	0.11	0.00	0.00	0.00	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.00	0.00	0.00	0.00	0.71	0.53	0.04
EI_Curiosity	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.26	0.00	0.01	0.00	0.00	0.00	0.02	0.38	0.17
EI_Independence	0.02	0.05	0.00	0.00	0.00	0.00	0.12	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.57	0.01	0.24	0.00	0.00	0.20	0.40	0.06
EI_Mastery	0.05	0.98	0.04	0.41	0.56	0.00	0.36	0.51	0.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.24	0.00	0.00	0.00	0.00	0.15	0.00	0.00
EI_Easy	0.16	0.79	0.02	0.99	0.40	0.00	0.30	0.38	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.68	0.00	0.00	0.00	0.00	0.80	0.04	0.01
EI_Pleasing																								

	CN_Security	CN_Accuracy	CN_Diligence	CN_Perseverance	CN_Tendency	Com_Mastery	MC_Learning	MC_Self-efficacy	MC_Self-regulation	EI_Confidence	EI_Independence	EI_Cultural	EI_Dependence on teacher	EI_Pleasant	MC_Self-achievement	MC_Orientation	NE_Neuroticism	NE_Openness	NE_Conscientiousness	Emotional Intelligence	Letter Sets	Figure Classification	GPA
EI_Dependence on teacher	0.00	0.00	0.11	0.03	0.00	0.00	0.00	0.00	0.08	0.00	0.26	0.00	0.00	0.00	0.00	0.17	0.00	0.18	0.29	0.01	0.16	0.02	0.00	0.61
MC_Self-achievement	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00	0.00	0.00	0.00	0.00	0.00	0.13	0.60	0.11
MC_Orientation	0.03	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.18	0.01	0.57	0.24	0.68	0.00	0.00	0.00	0.39	0.87	0.95	0.00	0.06	0.03	0.22
NE_Neuroticism	0.38	0.05	0.46	0.91	0.95	0.00	0.73	0.96	0.02	0.00	0.00	0.01	0.00	0.00	0.18	0.00	0.39	0.00	0.41	0.00	0.00	0.52	0.30	0.84
NE_Openness	0.03	0.00	0.04	0.33	0.12	0.02	0.03	0.04	0.00	0.00	0.00	0.24	0.00	0.00	0.29	0.00	0.87	0.41	0.00	0.04	0.00	0.22	0.10	0.91
NE_Conscientiousness	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.95	0.00	0.04	0.00	0.00	0.86	0.21	0.00
Emotional Intelligence	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.16	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.01	0.16
Letter Sets	0.24	0.10	0.01	0.23	0.00	0.02	0.55	0.55	0.02	0.71	0.02	0.20	0.15	0.80	0.02	0.13	0.06	0.52	0.22	0.86	0.03	0.00	0.00	0.24
Figure Classification	0.28	0.31	0.00	0.02	0.00	0.00	0.65	0.07	0.01	0.53	0.38	0.40	0.00	0.04	0.00	0.60	0.03	0.30	0.10	0.21	0.01	0.00	0.00	0.04
GPA	0.99	0.28	0.08	0.62	0.17	0.13	0.05	0.58	0.00	0.04	0.17	0.06	0.00	0.01	0.61	0.11	0.22	0.84	0.91	0.00	0.16	0.24	0.04	0.00

```
r <- corr_test$r
p <- corr_test$p

stars <- ifelse(p < .001, "***", ifelse(p < .01, "**", ifelse(p < .05, "*", "")))

corr_table <- matrix(
  paste0(round(r, 2), stars),
  nrow = nrow(r),
  dimnames = dimnames(r))

corr_table |> kable(digits = 2)
```

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence	CN_Meaning	M_Safety	M_Self-esteem	M_Belonging	EI_Dependence	EI_Independence	EI_Classroom	EI_Learning	EI_Pleasantness	EI_Teacher	MC_Achievement	MC_Self-actualization	NE_Neuroticism	NE_Extraversion	NE_Openness	Emotional Intelligence	Letter Set	Figure Classification	GPA
CN_Security	1***	0.66***	0.53***	0.58***	0.48***	0.51***	0.57***	0.59***	0.38***	0.08	0.22***	0.11*	0.1*	0.07	0.2***	0.27***	0.11**	0.11**	0.11**	0.23***	0.29***	-0.06	0.05
CN_Acceptance	0.66***	1***	0.55***	0.49***	0.47***	0.32***	0.45***	0.46***	0.43***	0.15***	0.26***	0.1*	0	0.01	0.18***	0.23***	0.08	0.1*	0.18***	0.2***	0.28***	-0.08	0.05	0.06
CN_Initiative	0.53***	0.55***	1***	0.7***	0.61***	0.36***	0.36***	0.59***	0.59***	0.35***	0.4***	0.26***	-0.1*	-0.11*	0.08	0.42***	0.19***	0.11**	0.25***	0.43***	-0.12*	-0.21***	0.09
CN_Competence	0.58***	0.49***	0.7***	1***	0.6***	0.44***	0.42***	0.72***	0.49***	0.27***	0.32***	0.24***	0	0.04	0.11*	0.45***	0.24***	0.01	0.05	0.28***	0.38***	-0.06	-0.11*	0.03
CN_Meaning	0.48***	0.47***	0.61***	0.6***	1***	0.42***	0.41***	0.58***	0.48***	0.32***	0.42***	0.23***	-0.03	-0.04	0.17***	0.38***	0.16***	0.08	0.3***	0.39***	-0.17*	-0.16***	0.07
M_Safety	0.51***	0.32***	0.36***	0.44***	0.42***	1***	0.4***	0.53***	0.31***	0.03	0.15***	0.15***	0.22***	0.15***	0.19***	0.19***	0.26***	0.18***	0.24***	0.15***	-0.12*	-0.15***	0.08
M_Belonging	0.57***	0.45***	0.36***	0.42***	0.41***	0.4***	1***	0.61***	0.42***	0.15***	0.25***	0.08	0.05	0.05	0.22***	0.38***	0.19***	0.02	0.11*	0.3***	0.41***	-0.03	0.02	0.1
M_Self-esteem	0.59***	0.46***	0.59***	0.72***	0.58***	0.53***	0.61***	1***	0.62***	0.23***	0.31***	0.17***	0.03	0.04	0.16***	0.48***	0.3***	0.1*	0.28***	0.46***	-0.03	-0.09	0.03

	CN_Security	CN_Acceptance	CN_Curiosity	CN_Tinkering	CN_Feeling	M_Becoming	M_Self-actualization	M_Self-esteem	M_Safety	EI_Challenge	EI_Curiosity	EI_Independence	EI_Easy work	EI_Pleasing	EI_Dependence on teacher	MC_Self-achievement	MC_Power	NE_Neuroticism	NE_Openness	NE_Emotional Intelligence	NE_Conscientiousness	Letter Sets	Figure Classification	GPA
M_Self-actualization	0.38***	0.43***	0.59***	0.49***	0.48***	0.31***	0.42***	0.62***		0.27***	0.36***	0.13***	-0.06	0.03	0.09	0.45***	0.18***	0.26***	0.19***	0.5***		-0.12*	-0.13**	-0.18***
EI_Challenge	0.08	0.15***	0.35***	0.27***	0.32***	0.03	0.15***	0.23***	0.27***		0.76***	0.59***	-0.69***	-0.45***	-0.14**	0.59***	0.07	0.22***	0.33***	0.44***	0.02	0.03	0.03	0.11*
EI_Curiosity	0.22***	0.26***	0.4***	0.32***	0.42***	0.15***	0.25***	0.31***	0.36***	0.76***		0.47***	-0.51***	-0.43***	0.06	0.5***	0.13**	0.24***	0.3***	0.51***	-0.11***	-0.04		0.07
EI_Independence	0.1	0.1	0.26***	0.24***	0.23***	0.15***	0.08	0.17***	0.13***	0.59***	0.47***		-0.41***	-0.26***	-0.31***	0.38***	0.03	0.06	0.26***	0.26***	0.06	0.04	0.04	0.1
EI_Easy work	0.1*		-0.1*	-0.04	-0.03	0.22***	0.05*	0.03		0.69***	0.51***	0.41***	-1***	-0.64***	-0.29***	0.42***	0.06	0.31***	-0.28***	-0.24***	-0.26***	-0.07	-0.15***	-0.16***
EI_Pleasing	0.07	0.01	-0.11*	0	-0.04	0.15***	0.05*	0.04		0.45***	0.43***	0.26***	-0.64***	-0.25***		0.22***	0.02	0.21***	-0.2***	-0.15***	-0.16***	-0.01	-0.11*	-0.14***
EI_Dependence on teacher	0.2***	0.18***	0.08	0.11*	0.17***	0.19***	0.22***	0.16***	0.09	-0.14**	0.06	0.31***	-0.29***	-0.25***		-0.07	0.16***	-0.14***	0.07	-0.05	-0.12**	-0.15**		0.03
MC_Self-achievement	0.26***	0.23***	0.42***	0.45***	0.38***	0.19***	0.38***	0.48***	0.45***	0.59***	0.5***	0.38***	-0.42***	-0.22***	-0.07	1***	0.16***	0.21***	0.45***	0.5***	0.08	-0.03		0.09
MC_Power	0.08	0.08	0.19***	0.24***	0.16***	0.26***	0.19***	0.18***	0.07	0.13***	0.03	0.06	0.02	0.16***	0.16***	1***	0.04	0.01	0.18***	-0.1	0.11*			0.07
NE_Neuroticism	0.1*	-0.04	0.01	0	0.18***	0.02	-0.12*	-0.25***	-0.22***	-0.14**		0.31***	-0.21***	-0.07	-0.15**	0.04	0.04***	0.04	-0.32***	-0.43***	0.03	0.05	-0.01	
NE_Openness	0.1*	0.18***	0.11***	0.05	0.08	-0.12*	0.1*	0.26***	0.22***	0.24***	0.06	-0.28***	-0.2***	-0.05		0.21***	0.01	0.04***	-0.1*	0.25***	0.06	0.08	0.01	

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence	CN_Meaning	M_Belongingness	M_Safety	Self_actualization	EI_Confidence	EI_Independence	EI_Dependence	EI_Easy_going	EI_Pleasantness	MC_Self_achievement	MC_Meticulousness	NE_Neuroticism	NE_Openness	Emotional Intelligence	Letter Sets	Figure Classification	GPA
NE_conscientiousness	-0.28***	-0.25***	-0.28***	-0.3***	-0.24***	-0.3***	-0.28***	-0.19***	-0.33***	-0.3***	-0.26***	-0.24***	-0.15***	0.14***	0.45***	-0.32***	-0.1***	1***	0.42***	0.01	0.21***
Emotional Intelligence	0.29***	0.28***	0.43***	0.38***	0.39***	0.15***	0.41***	0.46***	0.5***	0.44***	0.51***	0.26***	0.26***	0.07	0.5***	0.18***	0.25***	0.43***	1***	-0.11*	0.07
Letter Sets	-0.06	-0.08	-0.12**	-0.06	-0.17***	-0.12**	-0.03	-0.12*	-0.11*	-0.06	-0.07	-0.01	-0.12*	-0.08	-0.1	-0.03	0.06	0.01	-0.11*	1***	0.46***
Figure Classification	-0.05	-0.05	-0.21***	-0.11**	-0.16***	-0.15***	-0.02	-0.09	-0.13**	-0.04	-0.15***	-0.11**	-0.15***	-0.03	-0.11*	-0.05	0.08	-0.06	-0.13*	0.46***	0.11*
GPA	0.06	-0.09	-0.03	-0.07	-0.08	-0.1	-0.03	-0.18***	-0.11*	-0.07	-0.1	-0.16***	-0.14**	-0.03	-0.09	-0.07	0.01	-0.21***	-0.07	-0.06	1***

```
##cleaner
corr_table <- as.data.frame(corr_table)
corr_table
```

	CN_Security	CN_Acceptance	CN_Initiative	CN_Competence
CN_Security	1***	0.66***	0.53***	0.58***
CN_Acceptance	0.66***	1***	0.55***	0.49***
CN_Initiative	0.53***	0.55***	1***	0.7***
CN_Competence	0.58***	0.49***	0.7***	1***
CN_Meaning	0.48***	0.47***	0.61***	0.6***

M_Safety	0.51***	0.32***	0.36***	0.44***
M_Belonging	0.57***	0.45***	0.36***	0.42***
M_Self-esteem	0.59***	0.46***	0.59***	0.72***
M_Self-actualization	0.38***	0.43***	0.59***	0.49***
EI_Challenge	0.08	0.15**	0.35***	0.27***
EI_Curiosity	0.22***	0.26***	0.4***	0.32***
EI_Independence Mastery	0.11*	0.1	0.26***	0.24***
EI_Easy work	0.1*	0	-0.1*	-
0.04				
EI_Pleasing	0.07	0.01	-0.11*	0
EI_Dependence on teacher	0.2***	0.18***	0.08	0.11*
MC_Self-achievement	0.27***	0.23***	0.42***	0.45***
MC_Power	0.11*	0.08	0.19***	0.24***
NE_Neuroticism	0.11*	0.1*	-0.04	0.01
NE_Openness	0.11*	0.18***	0.11*	0.05
NE_Conscientiousness	0.23***	0.2***	0.25***	0.28***
Emotional Intelligence	0.29***	0.28***	0.43***	0.38***
Letter Sets	-0.06	-0.08	-0.12*	-
0.06				
Figure Classification	-0.05	-0.05	-0.21***	-
0.11*				
GPA	0	0.06	-0.09	-
0.03				

	CN_Meaning	M_Safety	M_Belonging	M_Self-esteem
CN_Security	0.48***	0.51***	0.57***	0.59***
CN_Acceptance	0.47***	0.32***	0.45***	0.46***
CN_Initiative	0.61***	0.36***	0.36***	0.59***
CN_Competence	0.6***	0.44***	0.42***	0.72***
CN_Meaning	1***	0.42***	0.41***	0.58***
M_Safety	0.42***	1***	0.4***	0.53***
M_Belonging	0.41***	0.4***	1***	0.61***
M_Self-esteem	0.58***	0.53***	0.61***	1***
M_Self-actualization	0.48***	0.31***	0.42***	0.62***
EI_Challenge	0.32***	0.03	0.15**	0.23***
EI_Curiosity	0.42***	0.15**	0.25***	0.31***
EI_Independence Mastery	0.23***	0.15**	0.08	0.17***
EI_Easy work	-0.03	0.22***	0.05	0.03
EI_Pleasing	-0.04	0.15**	0.05	0.04
EI_Dependence on teacher	0.17**	0.19***	0.22***	0.16**
MC_Self-achievement	0.38***	0.19***	0.38***	0.48***
MC_Power	0.16**	0.26***	0.19***	0.3***
NE_Neuroticism	0	0.18***	0.02	0
NE_Openness	0.08	-0.12*	0.11*	0.1*

NE_Conscientiousness	0.3***	0.24***	0.3***	0.28***
Emotional Intelligence	0.39***	0.15**	0.41***	0.46***
Letter Sets	-0.17***	-0.12*	0.03	-0.03
Figure Classification	-0.16**	-0.15**	-0.02	-0.09
GPA	-0.07	-0.08	0.1	-0.03
	M_Self-actualization		EI_Challenge	EI_Curiosity
CN_Security	0.38***		0.08	0.22***
CN_Acceptance	0.43***		0.15**	0.26***
CN_Initiative	0.59***		0.35***	0.4***
CN_Competence	0.49***		0.27***	0.32***
CN_Meaning	0.48***		0.32***	0.42***
M_Safety	0.31***		0.03	0.15**
M_Belonging	0.42***		0.15**	0.25***
M_Self-esteem	0.62***		0.23***	0.31***
M_Self-actualization	1***		0.27***	0.36***
EI_Challenge	0.27***		1***	0.76***
EI_Curiosity	0.36***		0.76***	1***
EI_Independence Mastery	0.13**		0.59***	0.47***
EI_Easy work	-0.06		-0.69***	-0.51***
EI_Pleasing	-0.03		-0.45***	-0.43***
EI_Dependence on teacher	0.09		-0.14**	0.06
MC_Self-achievement	0.45***		0.59***	0.5***
MC_Power	0.18***		0.07	0.13**
NE_Neuroticism	-0.12*		-0.25***	-0.22***
NE_Openness	0.26***		0.22***	0.24***
NE_Conscientiousness	0.19***		0.33***	0.3***
Emotional Intelligence	0.5***		0.44***	0.51***
Letter Sets	-0.12*		0.02	-0.11*
Figure Classification	-0.13**		0.03	-0.04
GPA	-0.18***		0.11*	0.07
	EI_Independence Mastery		EI_Easy work	EI_Pleasing
CN_Security	0.11*		0.1*	0.07
CN_Acceptance	0.1		0	0.01
CN_Initiative	0.26***		-0.1*	-0.11*
CN_Competence	0.24***		-0.04	0
CN_Meaning	0.23***		-0.03	-0.04
M_Safety	0.15**		0.22***	0.15**
M_Belonging	0.08		0.05	0.05
M_Self-esteem	0.17***		0.03	0.04
M_Self-actualization	0.13**		-0.06	-0.03
EI_Challenge	0.59***		-0.69***	-0.45***
EI_Curiosity	0.47***		-0.51***	-0.43***
EI_Independence Mastery	1***		-0.41***	-0.26***

EI_Easy work	-0.41***	1***	0.64***
EI_Pleasing	-0.26***	0.64***	1***
EI_Dependence on teacher	-0.31***	0.29***	0.25***
MC_Self-achievement	0.38***	-0.42***	-0.22***
MC_Power	0.03	0.06	0.02
NE_Neuroticism	-0.14**	0.31***	0.21***
NE_Openness	0.06	-0.28***	-0.2***
NE_Conscientiousness	0.26***	-0.24***	-0.15**
Emotional Intelligence	0.26***	-0.26***	-0.16**
Letter Sets	0.06	-0.07	0.01
Figure Classification	0.04	-0.15**	-0.11*
GPA	0.1	-0.16**	-0.14**

	EI_Dependence on teacher	MC_Self-achievement	MC_Power
CN_Security	0.2***	0.27***	0.11*
CN_Acceptance	0.18***	0.23***	0.08
CN_Initiative	0.08	0.42***	0.19***
CN_Competence	0.11*	0.45***	0.24***
CN_Meaning	0.17**	0.38***	0.16**
M_Safety	0.19***	0.19***	0.26***
M_Belonging	0.22***	0.38***	0.19***
M_Self-esteem	0.16**	0.48***	0.3***
M_Self-actualization	0.09	0.45***	0.18***
EI_Challenge	-0.14**	0.59***	0.07
EI_Curiosity	0.06	0.5***	0.13**
EI_Independence Mastery	-0.31***	0.38***	0.03
EI_Easy work	0.29***	-0.42***	0.06
EI_Pleasing	0.25***	-0.22***	0.02
EI_Dependence on teacher	1***	-0.07	0.16**
MC_Self-achievement	-0.07	1***	0.16**
MC_Power	0.16**	0.16**	1***
NE_Neuroticism	0.07	-0.15**	0.04
NE_Openness	-0.05	0.21***	0.01
NE_Conscientiousness	0.14**	0.45***	0
Emotional Intelligence	0.07	0.5***	0.18***
Letter Sets	-0.12*	0.08	-0.1
Figure Classification	-0.15**	-0.03	-0.11*
GPA	0.03	0.09	0.07

	NE_Neuroticism	NE_Openness	NE_Conscientiousness
CN_Security	0.11*	0.11*	0.23***
CN_Acceptance	0.1*	0.18***	0.2***
CN_Initiative	-0.04	0.11*	0.25***

CN_Competence	0.01	0.05	0.28***
CN_Meaning	0	0.08	0.3***
M_Safety	0.18***	-0.12*	0.24***
M_Belonging	0.02	0.11*	0.3***
M_Self-esteem	0	0.1*	0.28***
M_Self-actualization	-0.12*	0.26***	0.19***
EI_Challenge	-0.25***	0.22***	0.33***
EI_Curiosity	-0.22***	0.24***	0.3***
EI_Independence Mastery	-0.14**	0.06	0.26***
EI_Easy work	0.31***	-0.28***	-0.24***
EI_Pleasing	0.21***	-0.2***	-0.15**
EI_Dependence on teacher	0.07	-0.05	0.14**
MC_Self-achievement	-0.15**	0.21***	0.45***
MC_Power	0.04	0.01	0
NE_Neuroticism	1***	0.04	-0.32***
NE_Openness	0.04	1***	-0.1*
NE_Conscientiousness	-0.32***	-0.1*	1***
Emotional Intelligence	-0.43***	0.25***	0.42***
Letter Sets	0.03	0.06	0.01
Figure Classification	0.05	0.08	-0.06
GPA	-0.01	0.01	0.21***

	Emotional Intelligence	Letter Sets
CN_Security	0.29***	-0.06
CN_Acceptance	0.28***	-0.08
CN_Initiative	0.43***	-0.12*
CN_Competence	0.38***	-0.06
CN_Meaning	0.39***	-0.17***
M_Safety	0.15**	-0.12*
M_Belonging	0.41***	0.03
M_Self-esteem	0.46***	-0.03
M_Self-actualization	0.5***	-0.12*
EI_Challenge	0.44***	0.02
EI_Curiosity	0.51***	-0.11*
EI_Independence Mastery	0.26***	0.06
EI_Easy work	-0.26***	-0.07
EI_Pleasing	-0.16**	0.01
EI_Dependence on teacher	0.07	-0.12*
MC_Self-achievement	0.5***	0.08
MC_Power	0.18***	-0.1
NE_Neuroticism	-0.43***	0.03
NE_Openness	0.25***	0.06
NE_Conscientiousness	0.42***	0.01
Emotional Intelligence	1***	-0.11*

```
Letter Sets                              -0.11*        1***
Figure Classification                    -0.13*     0.46***
GPA                                       0.07        0.06
                         Figure Classification      GPA
CN_Security                              -0.05        0
CN_Acceptance                            -0.05     0.06
CN_Initiative                         -0.21***    -0.09
CN_Competence                            -0.11*    -0.03
CN_Meaning                               -0.16**    -0.07
M_Safety                                 -0.15**    -0.08
M_Belonging                              -0.02      0.1
M_Self-esteem                            -0.09    -0.03
M_Self-actualization                     -0.13** -0.18***
EI_Challenge                              0.03    0.11*
EI_Curiosity                             -0.04     0.07
EI_Independence Mastery                   0.04      0.1
EI_Easy work                             -0.15**  -0.16**
EI_Pleasing                              -0.11*   -0.14**
EI_Dependence on teacher                 -0.15**     0.03
MC_Self-achievement                      -0.03     0.09
MC_Power                                 -0.11*     0.07
NE_Neuroticism                            0.05    -0.01
NE_Openness                               0.08     0.01
NE_Conscientiousness                     -0.06  0.21***
Emotional Intelligence                   -0.13*     0.07
Letter Sets                           0.46***     0.06
Figure Classification                    1***    0.11*
GPA                                     0.11*     1***
```

```r
# Open a large, high-resolution image
png(
  "correlation_plot.png",
  width = 16, height = 14, units = "in",
  res = 300, pointsize = 12, bg = "white"
)

corrplot::corrplot(
  round(corr_test$r, 1),
  method = "color",
  type = "lower",          # shows only half of the matrix
  diag = FALSE,            # removes the 1.00 diagonal
```

```
  addCoef.col = "black",
  number.cex = 0.85,         # makes correlation numbers smaller

  number.font = 2,

  tl.cex = 0.8,             # makes variable names smaller
  tl.col = "black",
  tl.srt = 45,

  p.mat = corr_test$p,      # gives corrplot the p-values
  sig.level = 0.05,
  insig = "blank"           # hides nonsignificant correlations
)



# Finish saving the image
dev.off()
```

```
pdf
  2
```

CFA

First Order

```
# Define CFA model

CN_model1 <- '

Security =~ CN_1_1 + CN_1_2 + CN_1_3 + CN_1_4 + CN_1_5 + CN_1_6 + CN_1_7 + CN_1_8

Acceptance =~ CN_2_1 + CN_2_2 + CN_2_3 + CN_2_4 + CN_2_5 + CN_2_6 + CN_2_7 +CN_2_8

Initiative =~ CN_3_1 + CN_3_2 + CN_3_3 + CN_3_4 +CN_3_5 + CN_3_6 + CN_3_7 + CN_3_8

Competence =~ CN_4_1 + CN_4_2 + CN_4_3 + CN_4_4 +CN_4_5 + CN_4_6 + CN_4_7 + CN_4_8

Meaning =~ CN_5_1 + CN_5_2 + CN_5_3 + CN_5_4 + CN_5_5 + CN_5_6 + CN_5_7 + CN_5_8
```

```
'

fit_CN1 <- cfa(CN_model1, data = data, std.lv = TRUE,
               missing = "fiml",estimator = "MLR")


summary(fit_CN1,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
lavaan 0.7-2 ended normally after 127 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                       130

  Number of observations                           390
  Number of missing patterns                         2

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                              1687.560    1362.559
  Degrees of freedom                           730         730
  P-value (Chi-square)                          0.000       0.000
  Scaling correction factor                                 1.239
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                              8926.287    7071.385
  Degrees of freedom                           780         780
  P-value                                       0.000       0.000
  Scaling correction factor                                 1.262

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                   0.882       0.899
  Tucker-Lewis Index (TLI)                      0.874       0.893

  Robust Comparative Fit Index (CFI)                        0.903
  Robust Tucker-Lewis Index (TLI)                           0.896
```

```
Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -21998.892  -21998.892
  Scaling correction factor                                  1.219
      for the MLR correction
  Loglikelihood unrestricted model (H1)     -21155.112  -21155.112
  Scaling correction factor                                  1.236
      for the MLR correction

  Akaike (AIC)                               44257.785   44257.785
  Bayesian (BIC)                             44773.384   44773.384
  Sample-size adjusted Bayesian (SABIC)      44360.902   44360.902

Root Mean Square Error of Approximation:

  RMSEA                                          0.058       0.047
  90 Percent confidence interval - lower         0.054       0.044
  90 Percent confidence interval - upper         0.062       0.051
  P-value H_0: RMSEA <= 0.050                    0.000       0.912
  P-value H_0: RMSEA >= 0.080                    0.000       0.000

  Robust RMSEA                                               0.052
  90 Percent confidence interval - lower                     0.048
  90 Percent confidence interval - upper                     0.056
  P-value H_0: Robust RMSEA <= 0.050                         0.217
  P-value H_0: Robust RMSEA >= 0.080                         0.000

Standardized Root Mean Square Residual:

  SRMR                                           0.055       0.055

Goodness of Fit Index:

  Goodness of Fit Index (GFI)                    0.891
  90 Percent confidence interval - lower         0.878
  90 Percent confidence interval - upper         0.903

  Robust GFI                                                 0.910
  90 Percent confidence interval - lower                     0.896
  90 Percent confidence interval - upper                     0.923

Parameter Estimates:
```

```
  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  Security =~
    CN_1_1            0.806    0.054   15.040    0.000    0.806    0.694
    CN_1_2            0.896    0.065   13.678    0.000    0.896    0.678
    CN_1_3            0.755    0.058   13.085    0.000    0.755    0.673
    CN_1_4            0.769    0.076   10.057    0.000    0.769    0.505
    CN_1_5            0.499    0.054    9.155    0.000    0.499    0.507
    CN_1_6            0.586    0.063    9.252    0.000    0.586    0.551
    CN_1_7            0.762    0.070   10.909    0.000    0.762    0.559
    CN_1_8            0.833    0.060   13.969    0.000    0.833    0.685
  Acceptance =~
    CN_2_1            1.007    0.063   15.946    0.000    1.007    0.690
    CN_2_2            0.793    0.054   14.568    0.000    0.793    0.681
    CN_2_3            0.992    0.061   16.158    0.000    0.992    0.738
    CN_2_4            0.915    0.070   13.160    0.000    0.915    0.672
    CN_2_5            0.950    0.058   16.287    0.000    0.950    0.742
    CN_2_6            0.774    0.063   12.303    0.000    0.774    0.641
    CN_2_7            0.951    0.065   14.635    0.000    0.951    0.713
    CN_2_8            1.037    0.053   19.482    0.000    1.037    0.765
  Initiative =~
    CN_3_1            0.815    0.063   12.848    0.000    0.815    0.617
    CN_3_2            0.781    0.058   13.448    0.000    0.781    0.635
    CN_3_3            0.707    0.049   14.366    0.000    0.707    0.655
    CN_3_4            0.819    0.077   10.706    0.000    0.819    0.580
    CN_3_5            0.852    0.071   12.056    0.000    0.852    0.618
    CN_3_6            0.873    0.063   13.800    0.000    0.873    0.654
    CN_3_7            0.848    0.062   13.583    0.000    0.848    0.672
    CN_3_8            0.707    0.049   14.379    0.000    0.707    0.671
  Competence =~
    CN_4_1            0.802    0.041   19.503    0.000    0.802    0.778
    CN_4_2            0.770    0.039   19.791    0.000    0.770    0.765
    CN_4_3            0.782    0.041   19.146    0.000    0.782    0.758
    CN_4_4            0.776    0.043   17.899    0.000    0.776    0.717
    CN_4_5            0.784    0.045   17.615    0.000    0.784    0.720
    CN_4_6            0.773    0.049   15.657    0.000    0.773    0.705
    CN_4_7            0.678    0.045   15.210    0.000    0.678    0.661
    CN_4_8            0.668    0.051   13.049    0.000    0.668    0.634
```

```
  Meaning =~
    CN_5_1            1.172    0.064   18.325    0.000    1.172    0.811
    CN_5_2            1.220    0.060   20.362    0.000    1.220    0.800
    CN_5_3            1.209    0.059   20.322    0.000    1.209    0.790
    CN_5_4            0.961    0.063   15.309    0.000    0.961    0.732
    CN_5_5            1.102    0.066   16.685    0.000    1.102    0.713
    CN_5_6            1.097    0.067   16.259    0.000    1.097    0.737
    CN_5_7            1.045    0.064   16.224    0.000    1.045    0.776
    CN_5_8            1.271    0.063   20.037    0.000    1.271    0.821

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  Security ~~
    Acceptance        0.790    0.033   23.690    0.000    0.790    0.790
    Initiative        0.648    0.048   13.420    0.000    0.648    0.648
    Competence        0.670    0.039   17.259    0.000    0.670    0.670
    Meaning           0.542    0.047   11.490    0.000    0.542    0.542
  Acceptance ~~
    Initiative        0.636    0.049   13.055    0.000    0.636    0.636
    Competence        0.545    0.046   11.843    0.000    0.545    0.545
    Meaning           0.511    0.046   11.162    0.000    0.511    0.511
  Initiative ~~
    Competence        0.809    0.034   23.799    0.000    0.809    0.809
    Meaning           0.693    0.040   17.451    0.000    0.693    0.693
  Competence ~~
    Meaning           0.656    0.041   15.807    0.000    0.656    0.656

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .CN_1_1            5.569    0.059   94.737    0.000    5.569    4.797
   .CN_1_2            5.203    0.067   77.752    0.000    5.203    3.937
   .CN_1_3            5.792    0.057  102.033    0.000    5.792    5.167
   .CN_1_4            5.164    0.077   66.929    0.000    5.164    3.389
   .CN_1_5            6.072    0.050  121.798    0.000    6.072    6.168
   .CN_1_6            5.964    0.054  110.699    0.000    5.964    5.605
   .CN_1_7            5.264    0.069   76.310    0.000    5.264    3.864
   .CN_1_8            5.441    0.062   88.253    0.000    5.441    4.469
   .CN_2_1            4.592    0.074   62.160    0.000    4.592    3.148
   .CN_2_2            5.544    0.059   94.021    0.000    5.544    4.761
   .CN_2_3            5.151    0.068   75.713    0.000    5.151    3.834
   .CN_2_4            5.200    0.069   75.349    0.000    5.200    3.815
   .CN_2_5            5.361    0.065   82.611    0.000    5.361    4.183
   .CN_2_6            5.623    0.061   92.049    0.000    5.623    4.661
```

```
   .CN_2_7            5.123    0.068   75.838    0.000    5.123    3.840
   .CN_2_8            4.977    0.069   72.499    0.000    4.977    3.671
   .CN_3_1            5.067    0.067   75.761    0.000    5.067    3.836
   .CN_3_2            5.041    0.062   80.984    0.000    5.041    4.101
   .CN_3_3            5.554    0.055  101.614    0.000    5.554    5.145
   .CN_3_4            5.023    0.072   70.198    0.000    5.023    3.555
   .CN_3_5            4.905    0.070   70.253    0.000    4.905    3.557
   .CN_3_6            4.974    0.068   73.655    0.000    4.974    3.730
   .CN_3_7            5.318    0.064   83.255    0.000    5.318    4.216
   .CN_3_8            5.718    0.053  107.170    0.000    5.718    5.427
   .CN_4_1            5.685    0.052  108.932    0.000    5.685    5.516
   .CN_4_2            5.626    0.051  110.344    0.000    5.626    5.587
   .CN_4_3            5.815    0.052  111.371    0.000    5.815    5.639
   .CN_4_4            5.662    0.055  103.249    0.000    5.662    5.228
   .CN_4_5            5.605    0.055  101.571    0.000    5.605    5.143
   .CN_4_6            5.667    0.056  102.085    0.000    5.667    5.169
   .CN_4_7            5.833    0.052  112.331    0.000    5.833    5.688
   .CN_4_8            5.782    0.053  108.434    0.000    5.782    5.491
   .CN_5_1            5.262    0.073   71.864    0.000    5.262    3.639
   .CN_5_2            4.946    0.077   64.091    0.000    4.946    3.245
   .CN_5_3            5.156    0.077   66.538    0.000    5.156    3.369
   .CN_5_4            5.397    0.066   81.239    0.000    5.397    4.114
   .CN_5_5            5.069    0.078   64.789    0.000    5.069    3.281
   .CN_5_6            5.121    0.075   67.965    0.000    5.121    3.442
   .CN_5_7            5.372    0.068   78.786    0.000    5.372    3.989
   .CN_5_8            5.056    0.078   64.500    0.000    5.056    3.266

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .CN_1_1            0.699    0.070    9.935    0.000    0.699    0.519
   .CN_1_2            0.944    0.103    9.183    0.000    0.944    0.541
   .CN_1_3            0.688    0.075    9.127    0.000    0.688    0.547
   .CN_1_4            1.730    0.142   12.189    0.000    1.730    0.745
   .CN_1_5            0.720    0.070   10.275    0.000    0.720    0.743
   .CN_1_6            0.788    0.085    9.326    0.000    0.788    0.696
   .CN_1_7            1.275    0.108   11.801    0.000    1.275    0.687
   .CN_1_8            0.788    0.094    8.383    0.000    0.788    0.531
   .CN_2_1            1.115    0.093   12.045    0.000    1.115    0.524
   .CN_2_2            0.727    0.066   11.049    0.000    0.727    0.537
   .CN_2_3            0.822    0.105    7.812    0.000    0.822    0.455
   .CN_2_4            1.020    0.095   10.692    0.000    1.020    0.549
   .CN_2_5            0.739    0.079    9.416    0.000    0.739    0.450
   .CN_2_6            0.857    0.090    9.551    0.000    0.857    0.589
```

```
   .CN_2_7           0.876    0.115    7.613    0.000    0.876    0.492
   .CN_2_8           0.763    0.081    9.398    0.000    0.763    0.415
   .CN_3_1           1.080    0.095   11.405    0.000    1.080    0.619
   .CN_3_2           0.902    0.083   10.895    0.000    0.902    0.597
   .CN_3_3           0.666    0.062   10.695    0.000    0.666    0.571
   .CN_3_4           1.326    0.137    9.675    0.000    1.326    0.664
   .CN_3_5           1.175    0.122    9.664    0.000    1.175    0.618
   .CN_3_6           1.017    0.104    9.738    0.000    1.017    0.572
   .CN_3_7           0.872    0.076   11.434    0.000    0.872    0.548
   .CN_3_8           0.610    0.057   10.697    0.000    0.610    0.549
   .CN_4_1           0.419    0.037   11.330    0.000    0.419    0.395
   .CN_4_2           0.421    0.040   10.531    0.000    0.421    0.415
   .CN_4_3           0.452    0.043   10.486    0.000    0.452    0.425
   .CN_4_4           0.571    0.063    9.056    0.000    0.571    0.487
   .CN_4_5           0.573    0.051   11.278    0.000    0.573    0.482
   .CN_4_6           0.604    0.065    9.231    0.000    0.604    0.503
   .CN_4_7           0.593    0.063    9.480    0.000    0.593    0.563
   .CN_4_8           0.663    0.074    8.977    0.000    0.663    0.598
   .CN_5_1           0.716    0.084    8.481    0.000    0.716    0.342
   .CN_5_2           0.835    0.070   11.905    0.000    0.835    0.359
   .CN_5_3           0.880    0.091    9.685    0.000    0.880    0.376
   .CN_5_4           0.799    0.077   10.393    0.000    0.799    0.464
   .CN_5_5           1.174    0.122    9.616    0.000    1.174    0.492
   .CN_5_6           1.010    0.110    9.172    0.000    1.010    0.456
   .CN_5_7           0.721    0.079    9.100    0.000    0.721    0.398
   .CN_5_8           0.781    0.095    8.209    0.000    0.781    0.326
    Security         1.000                              1.000    1.000
    Acceptance       1.000                              1.000    1.000
    Initiative       1.000                              1.000    1.000
    Competence       1.000                              1.000    1.000
    Meaning          1.000                              1.000    1.000
```

```
#Main Fit Indice
fitMeasures(fit_CN1, c("cfi", "tli", "rmsea", "srmr"))
```

```
  cfi   tli rmsea  srmr
0.882 0.874 0.058 0.055
```

```
#Corelations among factors
lavInspect(fit_CN1, "cor.lv")
```

```
            Secrty Accptn Inittv Cmptnc Meanng
```

```
Security    1.000
Acceptance  0.790  1.000
Initiative  0.648  0.636  1.000
Competence  0.670  0.545  0.809  1.000
Meaning     0.542  0.511  0.693  0.656  1.000
```

```r
#modification indices
modificationIndices(fit_CN1,
                    sort. = TRUE,
                    minimum.value = 10)
```

```
           lhs op    rhs     mi    epc sepc.lv sepc.all sepc.nox
891      CN_3_5 ~~ CN_3_6 75.981  0.537   0.537    0.491    0.491
451      CN_1_5 ~~ CN_1_6 47.207  0.279   0.279    0.371    0.371
345      CN_1_2 ~~ CN_1_8 36.636  0.313   0.313    0.363    0.363
1071     CN_5_4 ~~ CN_5_5 33.102  0.312   0.312    0.322    0.322
154    Security =~ CN_3_6 26.368 -0.449  -0.449   -0.337   -0.337
343      CN_1_2 ~~ CN_1_6 26.343 -0.251  -0.251   -0.291   -0.291
380      CN_1_3 ~~ CN_1_6 24.631  0.207   0.207    0.281    0.281
151    Security =~ CN_3_3 22.487  0.336   0.336    0.311    0.311
851      CN_3_3 ~~ CN_3_5 22.445 -0.236  -0.236   -0.267   -0.267
550      CN_1_7 ~~ CN_5_6 21.680  0.288   0.288    0.254    0.254
342      CN_1_2 ~~ CN_1_5 20.775 -0.211  -0.211   -0.256   -0.256
179  Acceptance =~ CN_1_7 20.158  0.568   0.568    0.417    0.417
346      CN_1_2 ~~ CN_2_1 19.759  0.256   0.256    0.250    0.250
214  Initiative =~ CN_2_2 18.031  0.291   0.291    0.249    0.249
177  Acceptance =~ CN_1_5 17.893 -0.396  -0.396   -0.402   -0.402
186  Acceptance =~ CN_3_6 17.725 -0.344  -0.344   -0.258   -0.258
487      CN_1_6 ~~ CN_1_8 17.215 -0.186  -0.186   -0.236   -0.236
591      CN_2_1 ~~ CN_2_8 16.100  0.220   0.220    0.239    0.239
275     Meaning =~ CN_1_7 15.608  0.308   0.308    0.226    0.226
378      CN_1_3 ~~ CN_1_4 14.806  0.235   0.235    0.215    0.215
771      CN_2_7 ~~ CN_4_7 14.750  0.153   0.153    0.212    0.212
488      CN_1_6 ~~ CN_2_1 14.692 -0.196  -0.196   -0.209   -0.209
153    Security =~ CN_3_5 13.748 -0.343  -0.343   -0.249   -0.249
675      CN_2_4 ~~ CN_2_5 13.283  0.184   0.184    0.212    0.212
416      CN_1_4 ~~ CN_1_6 12.784  0.225   0.225    0.193    0.193
232  Initiative =~ CN_5_4 12.653  0.276   0.276    0.210    0.210
495      CN_1_6 ~~ CN_2_8 12.440 -0.153  -0.153   -0.198   -0.198
736      CN_2_6 ~~ CN_3_5 11.536 -0.185  -0.185   -0.185   -0.185
637      CN_2_2 ~~ CN_4_8 11.344 -0.127  -0.127   -0.183   -0.183
940      CN_3_7 ~~ CN_5_4 11.325  0.156   0.156    0.187    0.187
```

```
1040     CN_4_7 ~~ CN_5_4 11.123  0.125   0.125    0.182    0.182
350      CN_1_2 ~~ CN_2_5 10.902 -0.158  -0.158   -0.189   -0.189
379      CN_1_3 ~~ CN_1_5 10.662  0.129   0.129    0.183    0.183
1053     CN_5_1 ~~ CN_5_2 10.497  0.154   0.154    0.199    0.199
176  Acceptance =~ CN_1_4 10.059 -0.459  -0.459   -0.301   -0.301
278     Meaning =~ CN_2_2 10.024  0.181   0.181    0.156    0.156
211  Initiative =~ CN_1_7 10.001  0.299   0.299    0.219    0.219
```

```r
#Plot

# Open a high-resolution PNG
png(
  "CN1_CFA.png",
  width = 14, height = 22, units = "in",
  res = 300, pointsize = 12, bg = "white"
)

semPlot::semPaths(
  fit_CN1,
  what = "paths",
  whatLabels = "std",          # Standardized estimates
  layout = "tree2",
  rotation = 2,               # Factors left, items right

  nCharNodes = 0,              # Show full variable names
  nDigits = 2,
  sizeMan = 6,
  sizeMan2 = 2.5,
  sizeLat = 10,
  sizeLat2 = 5,
  edge.label.cex = 0.9,

  residuals = FALSE,
  intercepts = FALSE,
  exoCov = TRUE,               # Show factor correlations

  color = list(lat = "#D8E8F5", man = "white"),
  edge.color = "black",
  fade = FALSE,
  normalize = FALSE,
  mar = c(4, 8, 4, 4)
)
```

```
# Finish saving
dev.off()
```

```
pdf
  2
```

Second Order

```
# Define CFA model

CN_model2 <- '

Security =~ CN_1_1 + CN_1_2 + CN_1_3 + CN_1_4 + CN_1_5 + CN_1_6 + CN_1_7 + CN_1_8

Acceptance =~ CN_2_1 + CN_2_2 + CN_2_3 + CN_2_4 + CN_2_5 + CN_2_6 + CN_2_7 +CN_2_8

Initiative =~ CN_3_1 + CN_3_2 + CN_3_3 + CN_3_4 +CN_3_5 + CN_3_6 + CN_3_7 + CN_3_8

Competence =~ CN_4_1 + CN_4_2 + CN_4_3 + CN_4_4 +CN_4_5 + CN_4_6 + CN_4_7 + CN_4_8

Meaning =~ CN_5_1 + CN_5_2 + CN_5_3 + CN_5_4 + CN_5_5 + CN_5_6 + CN_5_7 + CN_5_8

CoreNeed =~ Security + Acceptance + Initiative + Competence + Meaning

'

fit_CN2 <- cfa(CN_model2, data = data, std.lv = TRUE,
               missing = "fiml",estimator = "MLR")


summary(fit_CN2,
        fit.measures = TRUE,
        standardized = TRUE)
```

```
lavaan 0.7-2 ended normally after 145 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                       125
```

```
  Number of observations                            390
  Number of missing patterns                          2

Model Test User Model:
                                                Standard      Scaled
  Test Statistic                                1769.113    1427.514
  Degrees of freedom                             735         735
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.239
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                                8926.287    7071.385
  Degrees of freedom                             780         780
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.262

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.873       0.890
  Tucker-Lewis Index (TLI)                       0.865       0.883

  Robust Comparative Fit Index (CFI)                         0.893
  Robust Tucker-Lewis Index (TLI)                            0.887

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -22039.669  -22039.669
  Scaling correction factor                                  1.214
      for the MLR correction
  Loglikelihood unrestricted model (H1)     -21155.112  -21155.112
  Scaling correction factor                                  1.236
      for the MLR correction

  Akaike (AIC)                               44329.338   44329.338
  Bayesian (BIC)                             44825.106   44825.106
  Sample-size adjusted Bayesian (SABIC)      44428.489   44428.489

Root Mean Square Error of Approximation:

  RMSEA                                          0.060       0.049
```

```
  90 Percent confidence interval - lower         0.056       0.046
  90 Percent confidence interval - upper         0.064       0.053
  P-value H_0: RMSEA <= 0.050                    0.000       0.654
  P-value H_0: RMSEA >= 0.080                    0.000       0.000

  Robust RMSEA                                               0.054
  90 Percent confidence interval - lower                     0.050
  90 Percent confidence interval - upper                     0.059
  P-value H_0: Robust RMSEA <= 0.050                         0.046
  P-value H_0: Robust RMSEA >= 0.080                         0.000

Standardized Root Mean Square Residual:

  SRMR                                           0.062       0.062

Goodness of Fit Index:

  Goodness of Fit Index (GFI)                    0.883
  90 Percent confidence interval - lower         0.870
  90 Percent confidence interval - upper         0.895

  Robust GFI                                                 0.902
  90 Percent confidence interval - lower                     0.888
  90 Percent confidence interval - upper                     0.916

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  Security =~
    CN_1_1            0.487    0.045   10.940    0.000    0.802    0.691
    CN_1_2            0.527    0.051   10.358    0.000    0.868    0.657
    CN_1_3            0.460    0.053    8.747    0.000    0.757    0.675
    CN_1_4            0.481    0.063    7.583    0.000    0.793    0.520
    CN_1_5            0.323    0.041    7.794    0.000    0.532    0.540
    CN_1_6            0.372    0.046    8.107    0.000    0.613    0.576
    CN_1_7            0.456    0.046    9.901    0.000    0.751    0.552
    CN_1_8            0.494    0.050    9.856    0.000    0.813    0.668
  Acceptance =~
```

CN_2_1	0.689	0.059	11.743	0.000	1.012	0.694
CN_2_2	0.545	0.045	12.043	0.000	0.801	0.688
CN_2_3	0.667	0.059	11.329	0.000	0.980	0.730
CN_2_4	0.618	0.060	10.231	0.000	0.908	0.667
CN_2_5	0.650	0.058	11.130	0.000	0.955	0.745
CN_2_6	0.526	0.048	10.914	0.000	0.772	0.640
CN_2_7	0.645	0.059	10.947	0.000	0.947	0.710
CN_2_8	0.708	0.051	13.784	0.000	1.040	0.767
Initiative =~						
CN_3_1	0.356	0.053	6.670	0.000	0.823	0.623
CN_3_2	0.339	0.052	6.501	0.000	0.783	0.637
CN_3_3	0.311	0.045	6.874	0.000	0.717	0.665
CN_3_4	0.356	0.051	6.998	0.000	0.821	0.581
CN_3_5	0.364	0.063	5.732	0.000	0.839	0.609
CN_3_6	0.370	0.061	6.053	0.000	0.854	0.640
CN_3_7	0.369	0.060	6.099	0.000	0.851	0.675
CN_3_8	0.306	0.047	6.541	0.000	0.705	0.669
Competence =~						
CN_4_1	0.412	0.044	9.280	0.000	0.799	0.776
CN_4_2	0.397	0.038	10.366	0.000	0.770	0.765
CN_4_3	0.403	0.044	9.154	0.000	0.782	0.758
CN_4_4	0.399	0.045	8.935	0.000	0.775	0.716
CN_4_5	0.404	0.045	8.954	0.000	0.783	0.719
CN_4_6	0.399	0.045	8.896	0.000	0.774	0.706
CN_4_7	0.351	0.040	8.793	0.000	0.682	0.665
CN_4_8	0.344	0.040	8.618	0.000	0.667	0.633
Meaning =~						
CN_5_1	0.787	0.059	13.268	0.000	1.174	0.812
CN_5_2	0.818	0.062	13.191	0.000	1.220	0.800
CN_5_3	0.810	0.057	14.222	0.000	1.207	0.789
CN_5_4	0.643	0.047	13.614	0.000	0.959	0.731
CN_5_5	0.740	0.051	14.523	0.000	1.103	0.714
CN_5_6	0.736	0.056	13.256	0.000	1.098	0.738
CN_5_7	0.701	0.061	11.547	0.000	1.045	0.776
CN_5_8	0.853	0.063	13.543	0.000	1.272	0.821
CoreNeed =~						
Security	1.308	0.167	7.815	0.000	0.794	0.794
Acceptance	1.077	0.127	8.454	0.000	0.733	0.733
Initiative	2.080	0.344	6.043	0.000	0.901	0.901
Competence	1.663	0.219	7.597	0.000	0.857	0.857
Meaning	1.106	0.108	10.241	0.000	0.742	0.742

Intercepts:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
.CN_1_1	5.569	0.059	94.737	0.000	5.569	4.797
.CN_1_2	5.203	0.067	77.752	0.000	5.203	3.937
.CN_1_3	5.792	0.057	102.033	0.000	5.792	5.167
.CN_1_4	5.164	0.077	66.929	0.000	5.164	3.389
.CN_1_5	6.072	0.050	121.798	0.000	6.072	6.167
.CN_1_6	5.964	0.054	110.700	0.000	5.964	5.605
.CN_1_7	5.264	0.069	76.310	0.000	5.264	3.864
.CN_1_8	5.441	0.062	88.253	0.000	5.441	4.469
.CN_2_1	4.592	0.074	62.160	0.000	4.592	3.148
.CN_2_2	5.544	0.059	94.021	0.000	5.544	4.761
.CN_2_3	5.151	0.068	75.713	0.000	5.151	3.834
.CN_2_4	5.200	0.069	75.349	0.000	5.200	3.815
.CN_2_5	5.361	0.065	82.615	0.000	5.361	4.183
.CN_2_6	5.623	0.061	92.049	0.000	5.623	4.661
.CN_2_7	5.123	0.068	75.838	0.000	5.123	3.840
.CN_2_8	4.977	0.069	72.498	0.000	4.977	3.671
.CN_3_1	5.067	0.067	75.761	0.000	5.067	3.836
.CN_3_2	5.041	0.062	80.984	0.000	5.041	4.101
.CN_3_3	5.554	0.055	101.614	0.000	5.554	5.145
.CN_3_4	5.023	0.072	70.198	0.000	5.023	3.555
.CN_3_5	4.905	0.070	70.253	0.000	4.905	3.557
.CN_3_6	4.974	0.068	73.655	0.000	4.974	3.730
.CN_3_7	5.318	0.064	83.255	0.000	5.318	4.216
.CN_3_8	5.718	0.053	107.170	0.000	5.718	5.427
.CN_4_1	5.685	0.052	108.932	0.000	5.685	5.516
.CN_4_2	5.626	0.051	110.344	0.000	5.626	5.587
.CN_4_3	5.815	0.052	111.371	0.000	5.815	5.639
.CN_4_4	5.662	0.055	103.249	0.000	5.662	5.228
.CN_4_5	5.605	0.055	101.571	0.000	5.605	5.143
.CN_4_6	5.667	0.056	102.085	0.000	5.667	5.169
.CN_4_7	5.833	0.052	112.332	0.000	5.833	5.688
.CN_4_8	5.782	0.053	108.434	0.000	5.782	5.491
.CN_5_1	5.262	0.073	71.864	0.000	5.262	3.639
.CN_5_2	4.946	0.077	64.091	0.000	4.946	3.245
.CN_5_3	5.156	0.077	66.538	0.000	5.156	3.369
.CN_5_4	5.397	0.066	81.239	0.000	5.397	4.114
.CN_5_5	5.069	0.078	64.789	0.000	5.069	3.281
.CN_5_6	5.121	0.075	67.965	0.000	5.121	3.442
.CN_5_7	5.372	0.068	78.786	0.000	5.372	3.989
.CN_5_8	5.056	0.078	64.500	0.000	5.056	3.266

Variances:

	Estimate	Std.Err	z-value	P(>|z|)	Std.lv	Std.all
.CN_1_1	0.705	0.071	9.891	0.000	0.705	0.523
.CN_1_2	0.993	0.106	9.329	0.000	0.993	0.569
.CN_1_3	0.685	0.078	8.823	0.000	0.685	0.545
.CN_1_4	1.693	0.140	12.083	0.000	1.693	0.729
.CN_1_5	0.686	0.070	9.833	0.000	0.686	0.708
.CN_1_6	0.756	0.088	8.637	0.000	0.756	0.668
.CN_1_7	1.291	0.111	11.608	0.000	1.291	0.696
.CN_1_8	0.821	0.099	8.279	0.000	0.821	0.554
.CN_2_1	1.105	0.093	11.847	0.000	1.105	0.519
.CN_2_2	0.714	0.065	10.911	0.000	0.714	0.527
.CN_2_3	0.844	0.110	7.673	0.000	0.844	0.468
.CN_2_4	1.032	0.096	10.767	0.000	1.032	0.556
.CN_2_5	0.730	0.077	9.440	0.000	0.730	0.444
.CN_2_6	0.859	0.091	9.430	0.000	0.859	0.590
.CN_2_7	0.882	0.118	7.464	0.000	0.882	0.496
.CN_2_8	0.757	0.082	9.262	0.000	0.757	0.412
.CN_3_1	1.067	0.093	11.443	0.000	1.067	0.612
.CN_3_2	0.898	0.083	10.820	0.000	0.898	0.594
.CN_3_3	0.650	0.061	10.604	0.000	0.650	0.558
.CN_3_4	1.323	0.139	9.541	0.000	1.323	0.662
.CN_3_5	1.197	0.122	9.830	0.000	1.197	0.630
.CN_3_6	1.050	0.106	9.946	0.000	1.050	0.590
.CN_3_7	0.867	0.076	11.333	0.000	0.867	0.545
.CN_3_8	0.613	0.057	10.774	0.000	0.613	0.552
.CN_4_1	0.423	0.037	11.360	0.000	0.423	0.398
.CN_4_2	0.421	0.039	10.690	0.000	0.421	0.415
.CN_4_3	0.452	0.044	10.352	0.000	0.452	0.425
.CN_4_4	0.572	0.066	8.724	0.000	0.572	0.488
.CN_4_5	0.574	0.051	11.242	0.000	0.574	0.483
.CN_4_6	0.603	0.065	9.235	0.000	0.603	0.501
.CN_4_7	0.587	0.062	9.412	0.000	0.587	0.558
.CN_4_8	0.664	0.074	8.997	0.000	0.664	0.599
.CN_5_1	0.712	0.083	8.550	0.000	0.712	0.341
.CN_5_2	0.835	0.070	11.899	0.000	0.835	0.360
.CN_5_3	0.884	0.091	9.686	0.000	0.884	0.378
.CN_5_4	0.801	0.077	10.446	0.000	0.801	0.465
.CN_5_5	1.171	0.121	9.646	0.000	1.171	0.491
.CN_5_6	1.009	0.110	9.152	0.000	1.009	0.456
.CN_5_7	0.722	0.079	9.124	0.000	0.722	0.398
.CN_5_8	0.780	0.095	8.172	0.000	0.780	0.325
.Security	1.000				0.369	0.369
.Acceptance	1.000				0.463	0.463

```
   .Initiative        1.000                               0.188    0.188
   .Competence        1.000                               0.266    0.266
   .Meaning           1.000                               0.450    0.450
    CoreNeed          1.000                               1.000    1.000
```

```
#Main Fit Indice
fitMeasures(fit_CN2, c("cfi", "tli", "rmsea", "srmr"))
```

```
  cfi   tli rmsea  srmr
0.873 0.865 0.060 0.062
```

```
#Corelations among factors
lavInspect(fit_CN2, "cor.lv")
```

```
           Secrty Accptn Inittv Cmptnc Meanng CoreNd
Security    1.000
Acceptance  0.582  1.000
Initiative  0.716  0.660  1.000
Competence  0.681  0.628  0.772  1.000
Meaning     0.589  0.543  0.668  0.636  1.000
CoreNeed    0.794  0.733  0.901  0.857  0.742  1.000
```

```
#modification indices
modificationIndices(fit_CN2,
                    sort. = TRUE,
                    minimum.value = 10)
```

```
            lhs op        rhs     mi    epc sepc.lv sepc.all sepc.nox
928       CN_3_5 ~~     CN_3_6 80.795  0.565   0.565    0.504    0.504
1118    Security ~~ Acceptance 71.140  0.733   0.733    0.733    0.733
382       CN_1_2 ~~     CN_1_8 46.759  0.371   0.371    0.410    0.410
488       CN_1_5 ~~     CN_1_6 40.308  0.253   0.253    0.351    0.351
151     Security =~     CN_3_6 33.649 -0.355  -0.585   -0.439   -0.439
1108      CN_5_4 ~~     CN_5_5 33.131  0.312   0.312    0.322    0.322
380       CN_1_2 ~~     CN_1_6 28.632 -0.267  -0.267   -0.308   -0.308
379       CN_1_2 ~~     CN_1_5 25.121 -0.235  -0.235   -0.285   -0.285
1124 Acceptance ~~ Competence 23.265 -0.461  -0.461   -0.461   -0.461
888       CN_3_3 ~~     CN_3_5 22.765 -0.238  -0.238   -0.270   -0.270
383       CN_1_2 ~~     CN_2_1 22.462  0.279   0.279    0.266    0.266
587       CN_1_7 ~~     CN_5_6 20.980  0.285   0.285    0.250    0.250
```

```
176  Acceptance =~     CN_1_7 20.791  0.261   0.383    0.281    0.281
417      CN_1_3 ~~     CN_1_6 20.614  0.190   0.190    0.264    0.264
150    Security =~     CN_3_5 20.278 -0.291  -0.479   -0.348   -0.348
524      CN_1_6 ~~     CN_1_8 20.227 -0.205  -0.205   -0.260   -0.260
148    Security =~     CN_3_3 19.364  0.214   0.353    0.327    0.327
1119   Security ~~ Initiative 19.163 -0.596  -0.596   -0.596   -0.596
211  Initiative =~     CN_2_2 16.007  0.124   0.286    0.245    0.245
272     Meaning =~     CN_1_7 15.980  0.222   0.330    0.242    0.242
171  Acceptance =~     CN_1_2 15.741  0.206   0.303    0.229    0.229
628      CN_2_1 ~~     CN_2_8 15.191  0.215   0.215    0.236    0.236
808      CN_2_7 ~~     CN_4_7 14.904  0.154   0.154    0.214    0.214
183  Acceptance =~     CN_3_6 13.976 -0.218  -0.320   -0.240   -0.240
712      CN_2_4 ~~     CN_2_5 13.959  0.191   0.191    0.219    0.219
1127 Initiative ~~ Competence 13.853  0.608   0.608    0.608    0.608
304    CoreNeed =~     CN_1_7 13.517  0.497   0.497    0.365    0.365
415      CN_1_3 ~~     CN_1_4 12.778  0.220   0.220    0.204    0.204
229  Initiative =~     CN_5_4 12.550  0.114   0.264    0.201    0.201
525      CN_1_6 ~~     CN_2_1 12.534 -0.178  -0.178   -0.195   -0.195
773      CN_2_6 ~~     CN_3_5 12.507 -0.195  -0.195   -0.192   -0.192
318    CoreNeed =~     CN_3_5 12.392 -0.833  -0.833   -0.604   -0.604
307    CoreNeed =~     CN_2_2 12.199  0.288   0.288    0.247    0.247
205  Initiative =~     CN_1_4 12.032 -0.186  -0.429   -0.282   -0.282
490      CN_1_5 ~~     CN_1_8 11.815 -0.148  -0.148   -0.197   -0.197
208  Initiative =~     CN_1_7 11.733  0.162   0.374    0.274    0.274
977      CN_3_7 ~~     CN_5_4 11.437  0.157   0.157    0.188    0.188
1077     CN_4_7 ~~     CN_5_4 11.313  0.126   0.126    0.184    0.184
161    Security =~     CN_4_8 11.195  0.145   0.238    0.226    0.226
532      CN_1_6 ~~     CN_2_8 11.015 -0.142  -0.142   -0.188   -0.188
471      CN_1_4 ~~     CN_3_8 10.553 -0.181  -0.181   -0.178   -0.178
316    CoreNeed =~     CN_3_3 10.516  0.586   0.586    0.543    0.543
959      CN_3_6 ~~     CN_5_3 10.456  0.175   0.175    0.182    0.182
1090     CN_5_1 ~~     CN_5_2 10.269  0.152   0.152    0.197    0.197
```

PCA

```r
# Select variables
data_PCA1 <- data[, c(
  "CN_S", "CN_A", "CN_I", "CN_C", "CN_M",
  "M_S", "M_B", "M_Se", "M_Sa",
  "EI_Ch", "EI_Cu", "EI_I", "EI_E", "EI_P", "EI_D",
```

```
  "MC_S", "MC_P",
  "NE_N", "NE_O", "NE_C", "EQ",
  "LS", "FC", "gpa"
)]


# Remove rows with missing values
data_PCA1 <- na.omit(data_PCA1)


# PCA
pca_result1 <- prcomp(data_PCA1, center = TRUE, scale. = TRUE)


# Eigenvalues
eigenvalues1 <- pca_result1$sdev^2
eigenvalues1
```

```
 [1] 6.6323449 3.6016259 1.6333038 1.4126361 1.2384872 1.1425682 1.0391148
 [8] 0.8257460 0.7610668 0.6612288 0.6409935 0.5548794 0.4933352 0.4566079
[15] 0.4375701 0.4080714 0.3515617 0.3369506 0.2746212 0.2670617 0.2532661
[22] 0.2329592 0.1975483 0.1464512
```

```
# Kaiser rule: number of components with eigenvalue > 1
kaiser_components1 <- which(eigenvalues1 > 1)
kaiser_components1
```

```
[1] 1 2 3 4 5 6 7
```

```
n_components1 <- length(kaiser_components1)
n_components1
```

```
[1] 7
```

```
# Percentage of variance explained
variance_explained1 <- eigenvalues1 / sum(eigenvalues1)

variance_explained1 * 100
```

```
 [1] 27.6347704 15.0067746  6.8054326  5.8859837  5.1603634  4.7607008
 [7]  4.3296452  3.4406085  3.1711116  2.7551199  2.6708061  2.3119976
[13]  2.0555635  1.9025331  1.8232088  1.7002973  1.4648403  1.4039609
[19]  1.1442548  1.1127570  1.0552753  0.9706634  0.8231178  0.6102132
```

```r
# Cumulative percentage of variance explained
cumulative_variance1 <- cumsum(variance_explained1)

cumulative_variance1 * 100
```

```
 [1]  27.63477  42.64155  49.44698  55.33296  60.49332  65.25403  69.58367
 [8]  73.02428  76.19539  78.95051  81.62132  83.93331  85.98888  87.89141
[15]  89.71462  91.41492  92.87976  94.28372  95.42797  96.54073  97.59601
[22]  98.56667  99.38979 100.00000
```

```r
# PCA with Varimax rotation
pca_varimax1 <- psych::principal(
  data_PCA1,
  nfactors = n_components1,
  rotate = "varimax"
)


# Rotated component loadings
print(pca_varimax1$loadings, cutoff = 0.30)
```

```
Loadings:
      RC1    RC2    RC6    RC3    RC5    RC4    RC7
CN_S   0.827
CN_A   0.745
CN_I   0.758
CN_C   0.783
CN_M   0.715
M_S    0.642                      -0.393
M_B    0.674
M_Se   0.807                                    0.305
M_Sa   0.644                       0.306
EI_Ch         0.845
EI_Cu  0.322  0.697
EI_I          0.714
```

```
EI_E         -0.820
EI_P         -0.728
EI_D         -0.422                        0.464
MC_S   0.440  0.483  0.357
MC_P                                              0.873
NE_N                -0.807
NE_O                                0.810
NE_C   0.319         0.551        -0.353  0.364
EQ     0.422         0.621
LS                          0.860
FC                          0.748
gpa                                       0.782

                 RC1   RC2   RC6   RC3   RC5   RC4   RC7
SS loadings    5.675 3.698 1.799 1.663 1.355 1.297 1.213
Proportion Var 0.236 0.154 0.075 0.069 0.056 0.054 0.051
Cumulative Var 0.236 0.391 0.466 0.535 0.591 0.645 0.696
```

```r
# Scree plot
scree1 <- data.frame(
  PC = 1:length(eigenvalues1),
  Eigenvalue = eigenvalues1,
  Variance = variance_explained1,
  CumulativeVariance = cumulative_variance1,
  Keep = ifelse(eigenvalues1 > 1, "Keep", "Drop")
)


ggplot(scree1, aes(x = PC)) +
  geom_line(aes(y = Eigenvalue)) +
  geom_point(aes(y = Eigenvalue, color = Keep), size = 3) +
  geom_hline(yintercept = 1, linetype = "dashed") +

  geom_text_repel(
    data = subset(scree1, Keep == "Keep"),
    aes(
      y = Eigenvalue,
      label = paste0(
        "PC", PC, "\n",
        round(CumulativeVariance * 100, 1), "%"
      )
    ),
```

```
  nudge_y = 0.2,
  size = 4
) +

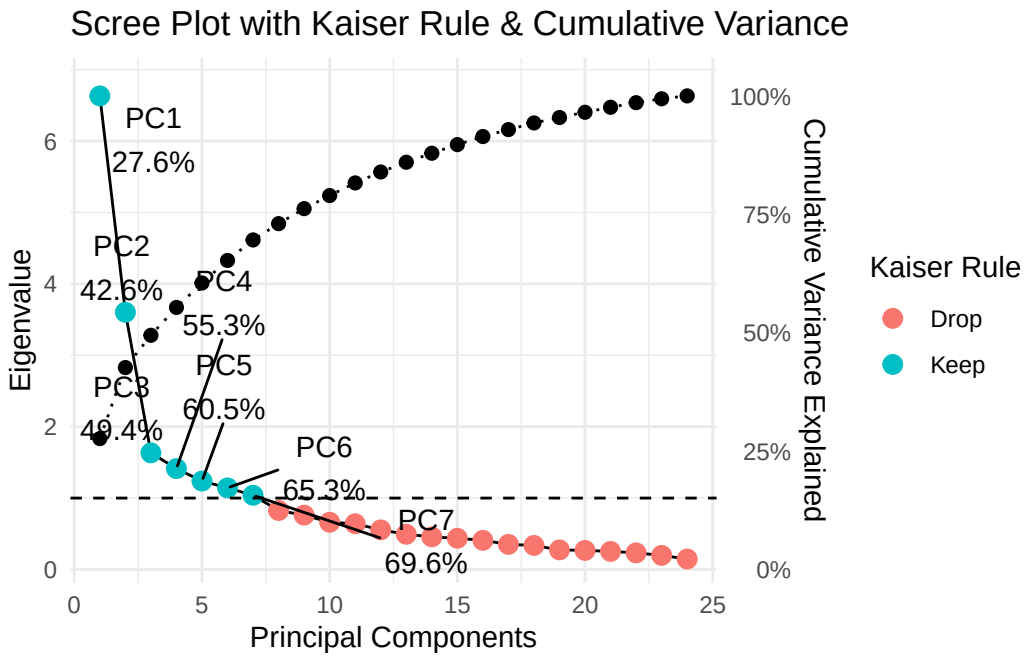
geom_line(
  aes(y = CumulativeVariance * max(eigenvalues1)),
  linetype = "dotted"
) +

geom_point(
  aes(y = CumulativeVariance * max(eigenvalues1)),
  size = 2
) +

scale_y_continuous(
  name = "Eigenvalue",
  sec.axis = sec_axis(
    ~ . / max(eigenvalues1),
    name = "Cumulative Variance Explained",
    labels = scales::percent
  )
) +

labs(
  title = "Scree Plot with Kaiser Rule & Cumulative Variance",
  x = "Principal Components",
  color = "Kaiser Rule"
) +

theme_minimal()
```


Scree Plot with Kaiser Rule & Cumulative Variance
Eigenvalue
Cumulative Variance Explained
Principal Components
PC1
27.6%
PC2
42.6%
PC3
49.4%
PC4
55.3%
PC5
60.5%
PC6
65.3%
PC7
69.6%
Kaiser Rule
Drop
Keep